

Point 1 briefing — Forward validation (Step 0)

Does the shallow-water formula recover HydroLight Rrs given true depth and true bottom reflectance?

Shared setup: structure and coefficients of the four formulas

Lee et al. (1998, 1999) give a semi-analytical shallow-water model in two structural forms: a single attenuation factor X , and a two-factor form with separate water-column and bottom terms. Equations ① and ② use the single- X form; Equations ③ and ④ use the two- X form. Each structure is run once with the original Lee coefficients and once with coefficients fitted on an independent set of 972 radiative-transfer simulations. The four combinations are the formulas validated here (Table C-1). Viewing is nadir, so $\cos(\theta_v^w) = 1$. The optical ratios are $u = b_b / (a + b_b)$ and $\kappa = a + b_b$.

Table C-1. The four formulas used in the forward validation.

Eq.	Structure	Du coefficients	Source
①	single- X	$DuC = 1.03 \sqrt{1 + 2.4u}$ $XC = XB$	Lee et al. (1998, 1999)
②	single- X	$DuC = C1 \sqrt{1 + C2 \cdot u}$ $C1 = 1.64670, C2 = 0.03234$	Fitted on 972 independent simulations
③	two- X	$DuC = 1.03 \sqrt{1 + 2.4u}$ $DuB = 1.04 \sqrt{1 + 5.4u}$	Lee et al. (1998, 1999)
④	two- X	$DuC = C1 \sqrt{1 + C2 \cdot u}, C1 = 0.50, C2 = 8$ $DuB = B1 \sqrt{1 + B2 \cdot u} = 1.61277 (B2 = 0)$	Fitted on 972 independent simulations

Table C-2. HydroLight conditions used to fit the empirical coefficients of Eqs. ② and ④ ($n = 972$). This set does not overlap the 288 scenes of Point 1.

Item	Levels
Water type	Classic Case-2
Solar zenith angle	10°, 30°, 50°
Chlorophyll-a	0.1, 0.3, 0.5 mg m ⁻³
TSM	0.1, 0.3, 0.5 mg m ⁻³
Bottom / depth / bands	Same family as Table 1; 972 cases in total
Independence from Point 1	Point 1 uses 20°/40°, Chl-a 0.2/0.4 mg m ⁻³ , TSM 0.2/0.4 mg m ⁻³ — no overlap

1. Forward validation — does the formula recover HydroLight Rrs?

Figure 1. Advisor comment (English rendering)

Step 0 — prior validation of the forward model. Depth inversion and bottom unmixing should not begin until the semi-analytical model is shown to be consistent with observed shallow-water Rrs. If known depth H and bottom reflectance $\rho(\lambda)$ are inserted and the modelled Rrs still fails to recover the observed spectrum, later errors cannot be attributed to IOP estimation, the attenuation coefficients, or the H - ρ trade-off. Instruction: using Lee et al.'s shallow-water forward model with known H and ρ , first confirm that the modelled Rrs reproduces both the magnitude and the shape of the observed shallow-water spectrum.

Problem

The long-term goal is to invert observed remote-sensing reflectance Rrs for water depth and bottom type. A mismatch with HydroLight shallow-water Rrs cannot, from inversion scores alone, be assigned to the shallow-water formula, to errors in the inherent optical properties (a , bb), or to a trade-off between depth and bottom reflectance. Earlier reports placed a forward calculation that used true depth in a retrieval-performance section; that layout could be misread as a depth-retrieval score, and it did not separate error by IOP source.

Purpose of this step

This step does not retrieve depth. True depth and true bottom reflectance are inserted into the shallow-water formula. The question is whether modelled Rrs reproduces HydroLight shallow-water Rrs in both magnitude and spectral shape under identical conditions. Holding depth and bottom at truth and changing only the source of a and bb partitions the residual into formula/Du error versus IOP-estimation error. That partition is required before later inversion error can be interpreted.

Procedure

For each scene, only true depth and true bottom reflectance are inserted. The four formulas (①–④) compute modelled Rrs at 443, 490, 560 and 665 nm, which is then compared with HydroLight shallow-water Rrs. Case A uses HydroLight truth for a and bb . Case B uses a and bb obtained by applying QAA to infinite-depth Rrs of the same water. Depth inversion and bottom mixing are not performed.

The validation set comprises 288 HydroLight radiative-transfer simulations of Classic Case-2 water: 6 bottom types $\times$ 6 depths $\times$ 2 solar zenith angles $\times$ 2 chlorophyll- a levels $\times$ 2 TSM levels (Table 1). Quantities inserted into the formula, and the comparison target, are listed in Table 2. Empirical coefficients of Eqs. ② and ④ were estimated under conditions independent of these 288 scenes and are not re-estimated here.

Metrics are RMSE and mean bias of the Rrs residual (model – HydroLight). Overall scores use 288 scenes $\times$ 4 bands = 1,152 points; band-wise scores use 288 points. In addition, to test whether the formula's DuC and DuB match HydroLight, Du inverted from HydroLight Rrs with true depth and bottom ("HydroLight-inverted Du") is compared with the formula's Du (Appendix A).

Table 1. HydroLight radiative-transfer conditions. The factorial design is $6 \times 6 \times 2 \times 2 \times 2 = 288$ cases.

Item	Setting
Water type	Classic Case-2
Bottom type	ooid sand, dark sediment, coral, brown algae, red algae, green algae (6)
Depth	0.5, 1, 2, 3, 4, 5 m
Solar zenith angle	20°, 40°
Wavelength	443, 490, 560, 665 nm
Chlorophyll-a	0.2, 0.4 mg m ⁻³
TSM	0.2, 0.4 mg m ⁻³
Temperature / salinity / sky	20 °C / 35 PSU / clear sky (cloud = 0)

Table 2. Experimental design of the forward validation. Depth is not retrieved in either case.

Case	Inserted into the formula	Compared against	Formulas	Data
A	True depth, true bottom ρ , HydroLight a and bb	HydroLight shallow Rrs	①–④	288
B	True depth, true bottom ρ , QAA a and bb	HydroLight shallow Rrs	①–④	288

Results

With true depth and true bottom reflectance prescribed, all four formulas recover the spectral shape of HydroLight shallow-water Rrs (Figures 2–9). Magnitude error (overall RMSE, Table 3) in Case A ranks as ② 1.49×10^{-3} , ④ 1.52×10^{-3} , ③ 1.68×10^{-3} , ① 2.44×10^{-3} . Eqs. ② and ④ (empirical coefficients) are smaller than Eqs. ① and ③ (original Lee coefficients). These numbers are not depth-retrieval scores.

When the IOPs are taken from QAA (Case B), RMSE of ① and ③ rises to 3.11×10^{-3} and 2.51×10^{-3} . Under the same switch, ② and ④ become 1.36×10^{-3} and 1.45×10^{-3} , similar to or slightly smaller than Case A. The extra error in ① and ③ is therefore due mainly to IOP estimation, whereas ② and ④ already have a smaller formula error and are little affected by the QAA switch. Mean bias is positive for ① and ③ (the model is brighter than HydroLight) and near zero or weakly negative for ② and ④. The Du comparison (Appendix A) shows that original Lee Du disagrees with the HydroLight-inverted value: in the single-X form (①), DuC is smaller than the inverted value; in the two-X form (③), the water-column term DuC is too large and the bottom term DuB is too small. Either mismatch brightens modelled Rrs, confirming that the positive bias of ① and ③ originates in the Du coefficients.

Eq.	Structure / coefficients	Case A RMSE	Case A Bias	Case B RMSE	Case B Bias	Interpretation
①	single-X, Lee	2.44	+1.37	3.11	+1.68	Rises in Case B. Extra error is IOP
②	single-X, empirical	1.49	-0.10	1.36	+0.18	Smallest. Insensitive to the QAA switch
③	two-X, Lee	1.68	+0.78	2.51	+1.08	Rises in Case B. Extra error is IOP
④	two-X, empirical	1.52	-0.68	1.45	-0.40	Similar to ②. Insensitive to the QAA switch

Table 3. Overall scores ($\times 10^{-3}$; 288 scenes $\times$ 4 bands = 1,152 points). RMSE = sqrt of the mean squared residual (model Rrs – HydroLight Rrs); Bias = mean(model – HydroLight). Positive bias means the model is brighter than HydroLight. Orange background = Case B. These are not depth-retrieval scores.

Conclusion

The Step 0 question is this: with true depth and true bottom reflectance inserted, does modelled R_{rs} recover both the shape and the magnitude of HydroLight shallow-water R_{rs} ?

Shape is recovered by all four formulas (Figures 2–9: the 560 nm peak and the 665 nm drop match HydroLight). Magnitude is judged on Case A (HydroLight-true a , b_b). Overall RMSE and bias ($\times 10^{-3}$, Table 3, 1,152 points) are ② 1.49 (bias -0.10), ④ 1.52 (-0.68), ③ 1.68 (+0.78), ① 2.44 (+1.37). Only Eq. ② has near-zero bias and the smallest RMSE, so both shape and magnitude match. Eqs. ① and ③ match the shape but are biased bright, so magnitude does not match; the cause is original Lee Du disagreeing with the HydroLight-inverted value (Appendix A).

The formula to adopt is Eq. ② (single-X, empirical coefficients). Use Eq. ④ if a two-X structure is required. Do not use Eqs. ① or ③ as the working formula.

Later depth and bottom inversion proceeds with Eq. ②. The residual that remains in Case A is formula/Du. When IOPs are taken from QAA (Case B), RMSE rises only for ① (2.44→3.11) and ③ (1.68→2.51), while ② stays essentially unchanged (1.49→1.36). Later error is therefore partitioned into formula error versus QAA error.

A. Spectral shape — modelled R_{rs} vs HydroLight R_{rs} (one formula, one case per panel)

The left column is Case A and the right column is Case B. Each panel pools 288 simulations by wavelength and plots the mean and the median. Where blue (model) and red (HydroLight) overlap, shape and magnitude agree. Depth is not retrieved.

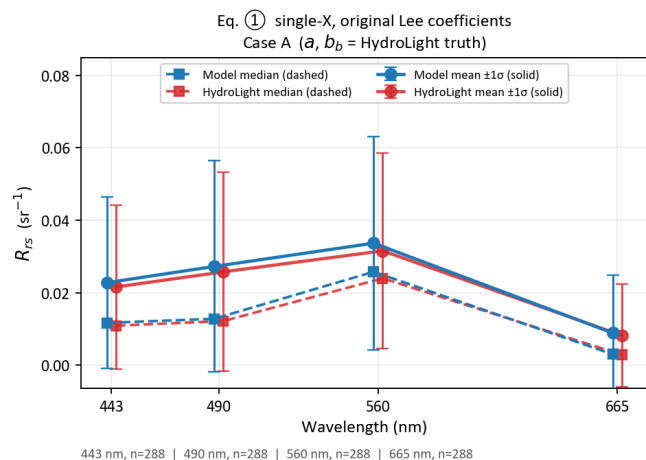


Figure 2. Eq. ① single-X, original Lee coefficients, Case A (a , b_b = HydroLight truth). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

Reading (Eq. ① (single-X, Lee)). Red and blue follow the same shape (a peak at 560 nm and a drop at 665 nm), but blue (model) lies above red. Mean bias is $+1.37 \times 10^{-3}$ (A) $\rightarrow +1.68 \times 10^{-3}$ (B), so the gap widens in Case B. The original-Lee single-X formula is biased bright.

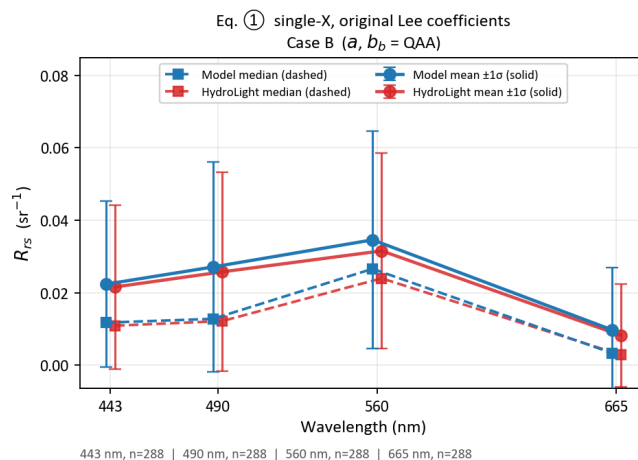


Figure 3. Eq. ① single-X, original Lee coefficients, Case B (a , b_b = QAA). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

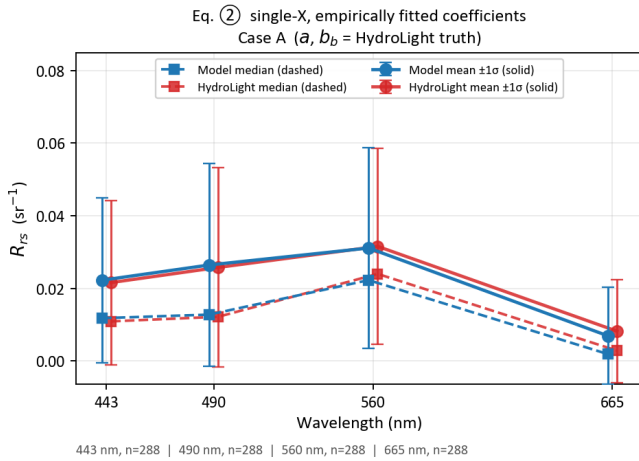


Figure 4. Eq. ② single-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

Reading (Eq. ② (single-X, empirical)). In both Case A and Case B, red and blue nearly overlap in mean, median and $\pm 1\sigma$ width. Mean bias is -0.10×10^{-3} (A) and $+0.18 \times 10^{-3}$ (B). This is the best spectral reproduction of the four formulas.

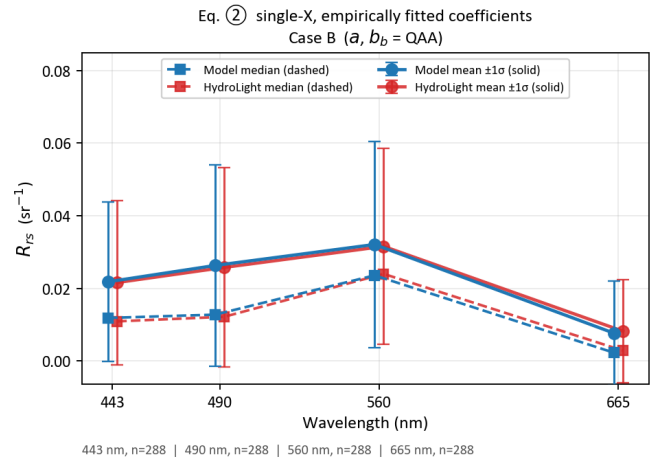


Figure 5. Eq. ② single-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

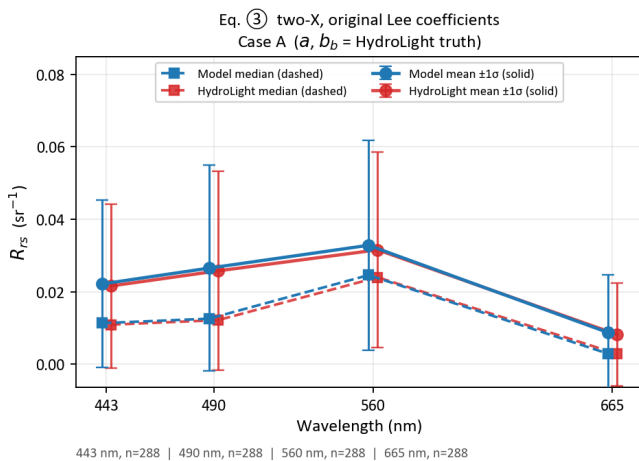


Figure 6. Eq. ③ two-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

Reading (Eq. ③ (two-X, Lee)). The shape is correct, but blue lies above red, and the 560–665 nm gap grows in Case B as in Eq. ①. Mean bias is $+0.78 \times 10^{-3}$ (A) $\rightarrow +1.08 \times 10^{-3}$ (B). Splitting X into two terms does not by itself stop QAA IOP error.

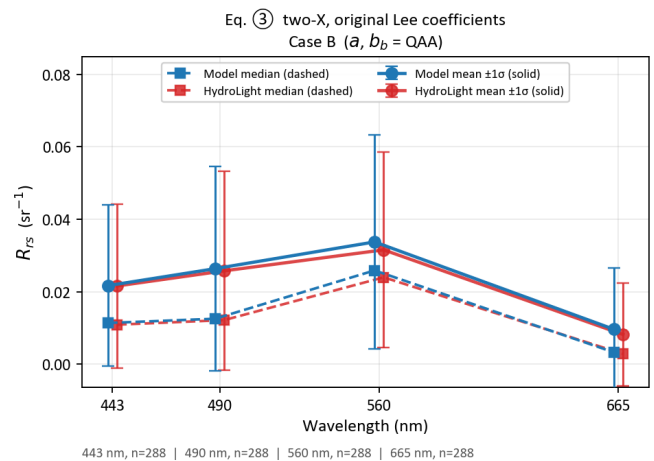


Figure 7. Eq. ③ two-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overlap.

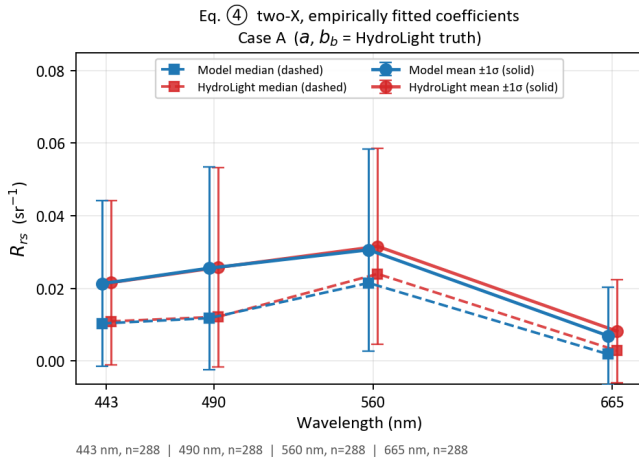


Figure 8. Eq. ④ two-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

Reading (Eq. ④ (two-X, empirical)). The two colours overlap much as in Eq. ②. Blue lies slightly below red (negative bias, -0.68×10^{-3} in A, -0.40×10^{-3} in B). The shape is preserved under the Case B switch.

B. Magnitude — band-wise Rrs RMSE (one formula, one case per panel)

Each bar is RMSE at one wavelength. The whisker is the standard deviation σ of the residual (model – HydroLight) and shows how much the error varies across scenes. The y-axis range is auto-scaled per panel, so compare magnitudes using the numbers on the bars ($\times 10^3$) and Table 4.

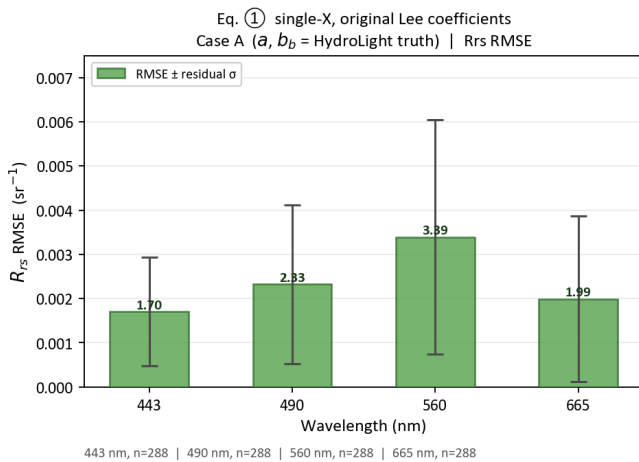


Figure 10. Band-wise Rrs RMSE for Eq. ① single-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ① (single-X, Lee)). In Case A, 560 nm is largest (3.39×10^{-3}). In Case B, 560 and 665 nm rise further to 4.65 and 3.43×10^{-3} , while 443 nm decreases. The taller bars are the effect of QAA a, b_b .

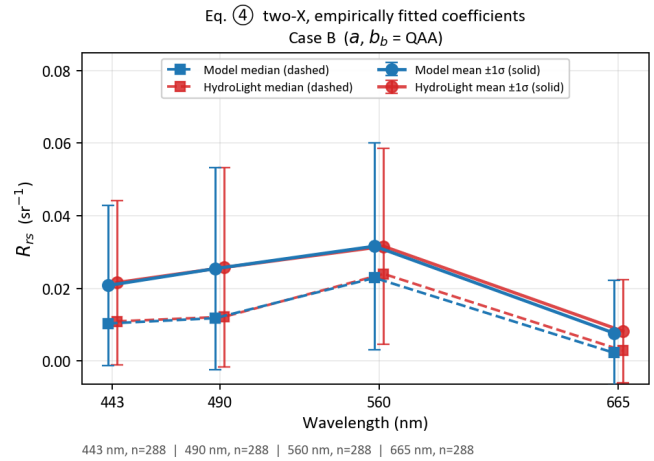


Figure 9. Eq. ④ two-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; R_{rs} (sr^{-1}) on the y-axis. Blue = model; red = HydroLight. Solid line with circles = mean of 288 scenes; vertical bars = $\pm 1\sigma$; dashed line with squares = median. The two traces are offset by ± 2 nm so they do not overplot.

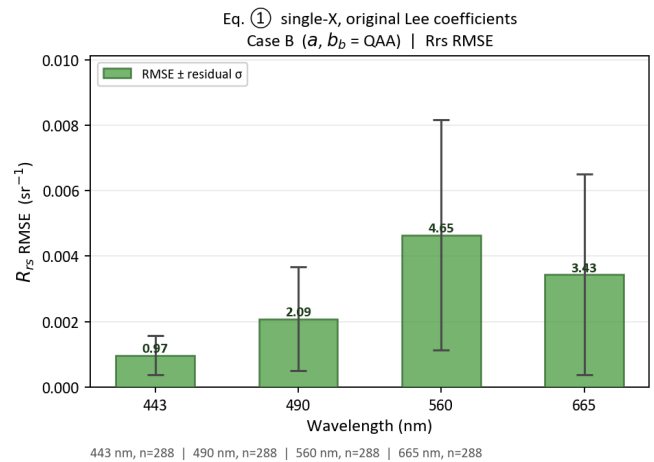


Figure 11. Band-wise Rrs RMSE for Eq. ① single-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

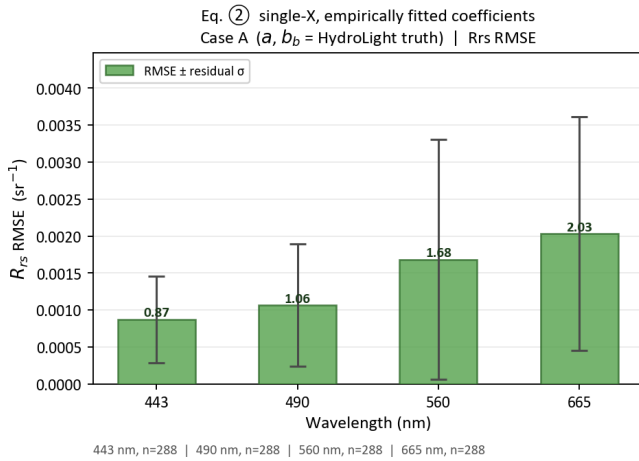


Figure 12. Band-wise Rrs RMSE for Eq. ② single-X, empirically fitted coefficients, Case A (a , b_b = HydroLight truth). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ② (single-X, empirical)). All four bands are similar in Case A and Case B ($0.9\text{--}2.0 \times 10^{-3}$). At 665 nm, Case B is actually smaller ($2.03 \rightarrow 1.15 \times 10^{-3}$). Formula error is already small, so the QAA switch barely moves the result.

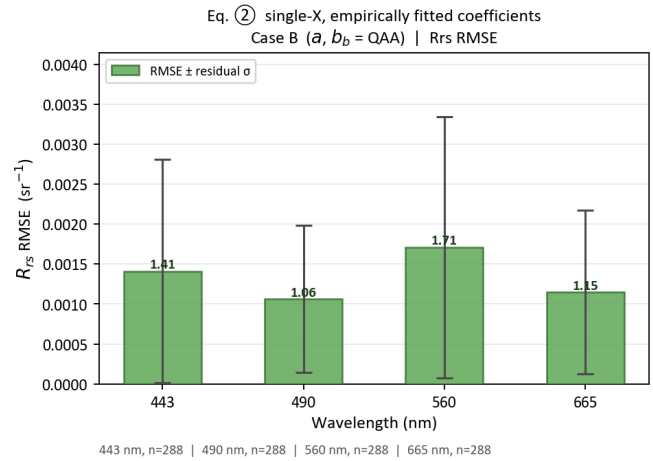


Figure 13. Band-wise Rrs RMSE for Eq. ② single-X, empirically fitted coefficients, Case B (a , b_b = QAA). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

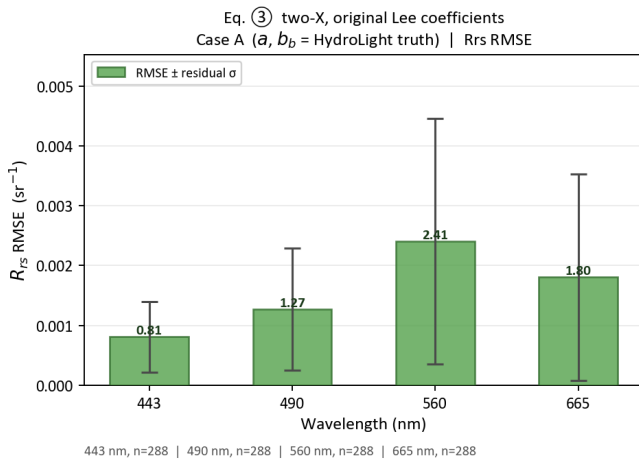


Figure 14. Band-wise Rrs RMSE for Eq. ③ two-X, original Lee coefficients, Case A (a , b_b = HydroLight truth). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ③ (two-X, Lee)). Case A is lower than Eq. ① (2.41×10^{-3} at 560 nm), but Case B again raises 560 and 665 nm to 3.55 and 3.23×10^{-3} . A two-X structure remains sensitive to IOP error if the original Lee coefficients are used.

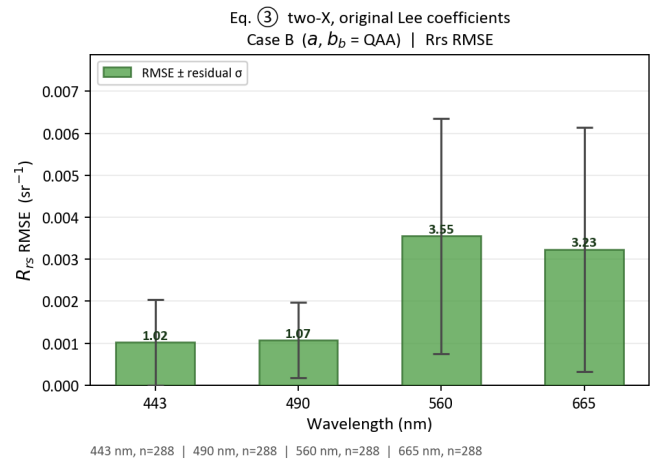


Figure 15. Band-wise Rrs RMSE for Eq. ③ two-X, original Lee coefficients, Case B (a , b_b = QAA). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

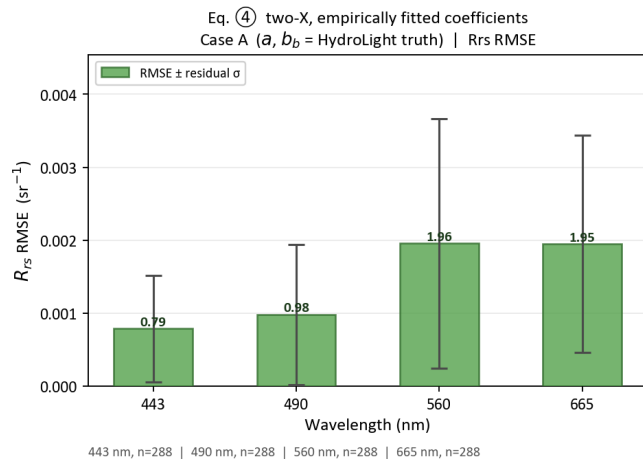


Figure 16. Band-wise Rrs RMSE for Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

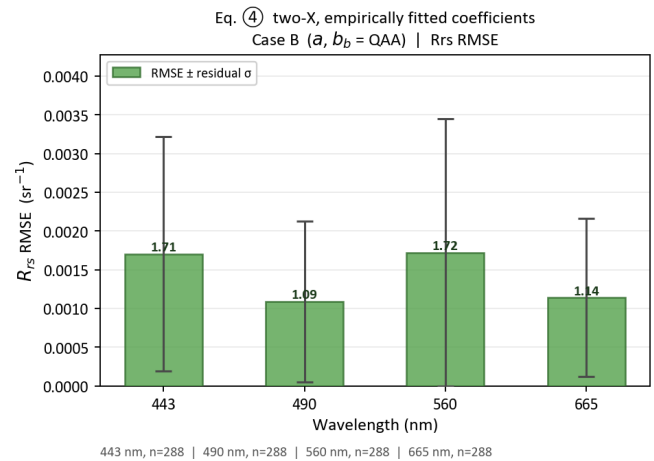


Figure 17. Band-wise Rrs RMSE for Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA). Wavelength (nm) on the x-axis; RMSE (sr^{-1}) on the y-axis. Green bars = RMSE (numbers on bars = $\times 10^3$); black whiskers = residual standard deviation $\pm \sigma$. $n = 288$ per band. The y-axis range differs by panel.

Reading (Eq. ④ (two-X, empirical)). The level matches Eq. ②. In Case B only 443 nm increases ($0.79 \rightarrow 1.71 \times 10^{-3}$) while 665 nm decreases ($1.95 \rightarrow 1.14 \times 10^{-3}$). Overall, the formula is insensitive to the QAA switch.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
($\times 10^{-3}$)	443	490	560	665	443	490	560	665
①	1.70	2.33	3.39	1.99	0.97	2.09	4.65	3.43
②	0.87	1.06	1.68	2.03	1.41	1.06	1.71	1.15
③	0.81	1.27	2.41	1.80	1.02	1.07	3.55	3.23
④	0.79	0.98	1.96	1.95	1.71	1.09	1.72	1.14

Table 4. Band-wise Rrs RMSE ($\times 10^{-3}$). Matches the bar heights in Figures 10–17. $n = 288$ per cell. Orange background = Case B.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
($\times 10^{-3}$)	443	490	560	665	443	490	560	665
①	+1.19	+1.48	+2.12	+0.68	+0.77	+1.35	+3.04	+1.56
②	+0.65	+0.67	-0.45	-1.28	+0.21	+0.54	+0.52	-0.53
③	+0.55	+0.75	+1.26	+0.55	+0.09	+0.60	+2.19	+1.43
④	-0.30	-0.20	-0.96	-1.26	-0.80	-0.33	+0.04	-0.51

Table 5. Band-wise mean bias ($\times 10^{-3}$) = mean(model – HydroLight). Positive = the model is brighter. ① and ③ are positive at all bands (larger at 560 and 665 nm in Case B). ② and ④ have small absolute values and are negative at 665 nm.

Eq.	443 nm	490 nm	560 nm	665 nm	Overall
①	-0.73	-0.24	+1.25	+1.44	+0.67
②	+0.54	+0.00	+0.03	-0.88	-0.13
③	+0.22	-0.20	+1.15	+1.43	+0.83
④	+0.92	+0.11	-0.23	-0.80	-0.07

Table 6. RMSE(Case B) – RMSE(Case A) ($\times 10^{-3}$) = right minus left in Table 4. Positive = error increases when a and b_b are taken from QAA. Depth and bottom are identical in the two cases, so this difference is solely the effect of IOP source. ① and ③ are +1.1 to +1.4 at 560 and 665 nm; ② and ④ are smaller and negative at 665 nm.

Summary. Shape is recovered by all four formulas; magnitude matches only for Eq. ② (Table 3). The formula to adopt is ②. The bright bias of ① and ③ originates in the Du coefficients (Appendix A).

Appendix A. Du comparison — formula DuC, DuB versus HydroLight-inverted Du

Tables 3–6 and Figures 2–17 already support the main-text conclusion. This appendix is supplementary evidence that the Rrs residual is rooted in the Du coefficients.

HydroLight-inverted Du is obtained by inserting HydroLight Rrs, true depth, true bottom reflectance and the case's a and b_b , then solving the formula for X , and setting $Du = X/\kappa - 1/\cos(\theta_w)$. It is the Du that would be required to match HydroLight exactly. Single- X formulas (①, ②) have $Du_C = Du_B$, so only Du_C is plotted. Two- X formulas (③, ④) plot the water-column term Du_C and the bottom term Du_B separately. Inversion fails when the argument of the logarithm leaves $(0, 1)$, so band-wise n can be smaller than 288 (shown under each panel); bands with $n < 30$ are omitted. Inverted values have a long tail, so a few scenes pull the mean; spectral plots in this appendix therefore use the median and the 25–75% interval.

DuC (water-column term) — spectra

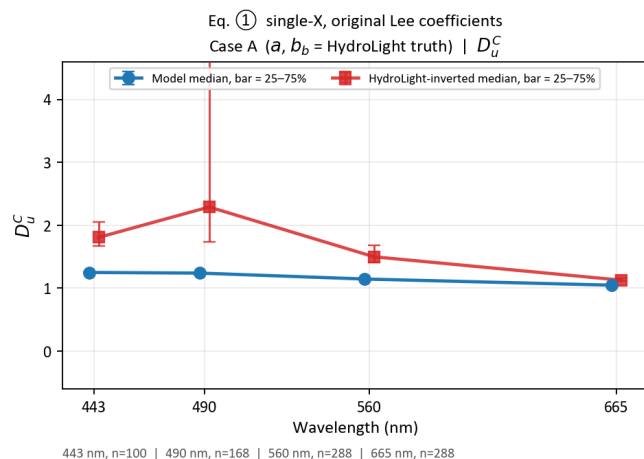


Figure A-1. Eq. ① single-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

Reading (Eq. ① (single-X, Lee)). Original Lee DuC (blue) is nearly flat at 1.05–1.25 across wavelength. HydroLight-inverted values (red) have medians 1.81 (443 nm) $\rightarrow$ 1.13 (665 nm), larger than blue and more dispersed. n is small at 443 and 490 nm (weak bottom signal at short wavelengths); treat those bands as indicative only. A single X must fit water column and bottom together, so the required Du becomes large. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

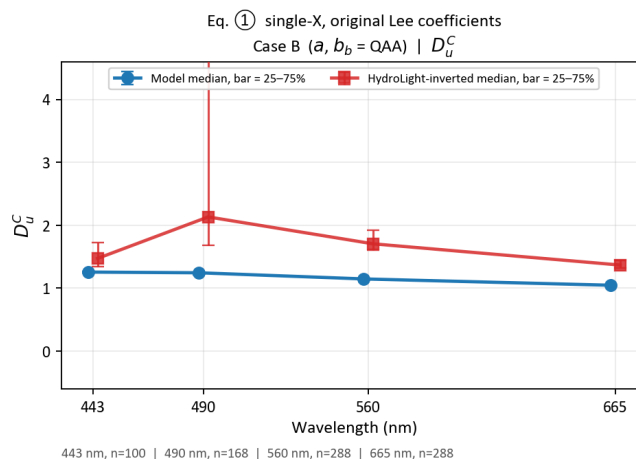


Figure A-2. Eq. ① single-X, original Lee coefficients, Case B (a, b_b = QAA): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

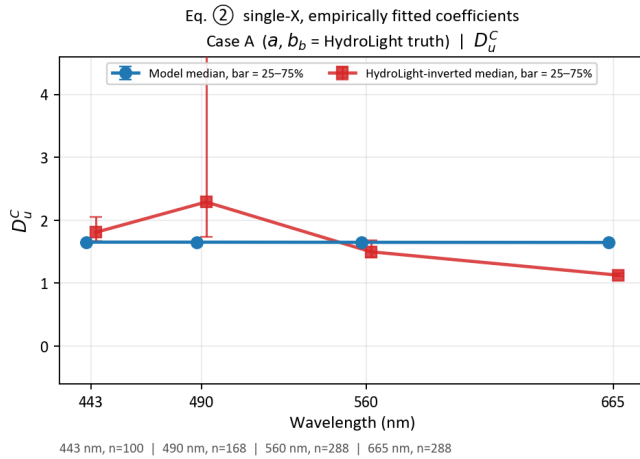


Figure A-3. Eq. ② single-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

Reading (Eq. ② (single-X, empirical)). Empirical DuC (blue) is nearly a constant 1.65. Inverted medians 1.50 (560 nm) and 1.13 (665 nm) lie closer to it than to Eq. ①. That is why the Rrs bias of Eq. ② is near zero. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

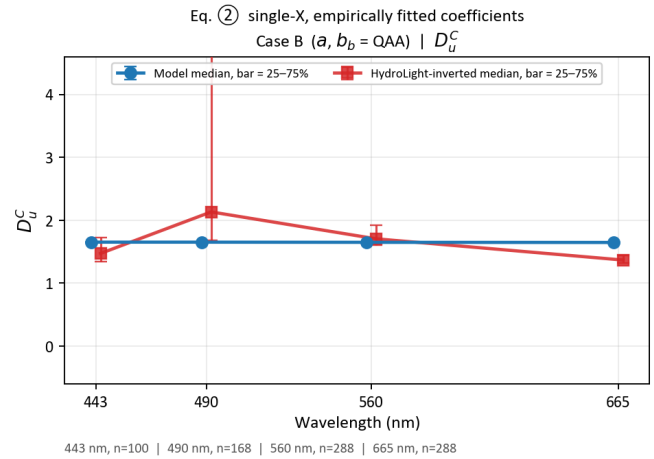


Figure A-4. Eq. ② single-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

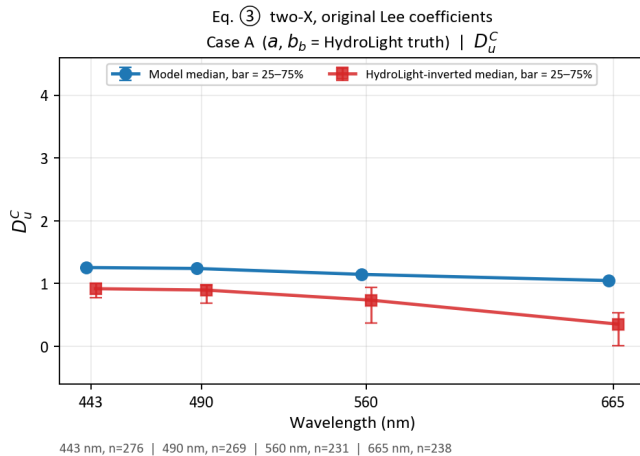


Figure A-5. Eq. ③ two-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

Reading (Eq. ③ (two-X, Lee)). In the two-X form, inverted water-column DuC (red) has medians 0.92 (443 nm) $\rightarrow$ 0.35 (665 nm), smaller than the Lee values 1.25–1.05 (blue). Original Lee coefficients overstate water-column attenuation, so modelled Rrs is too bright (positive bias of Eq. ③). In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

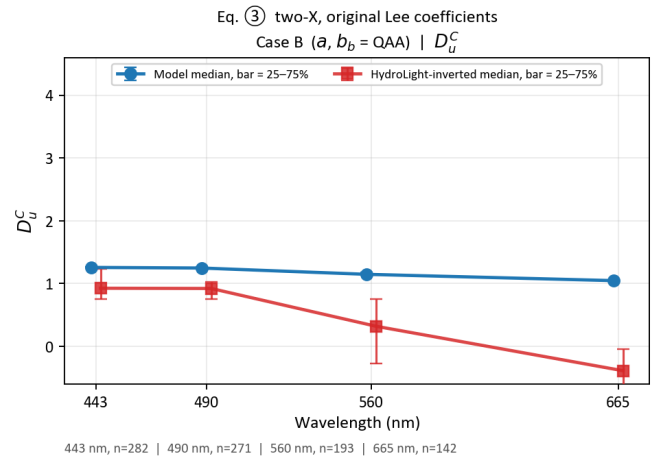


Figure A-6. Eq. ③ two-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

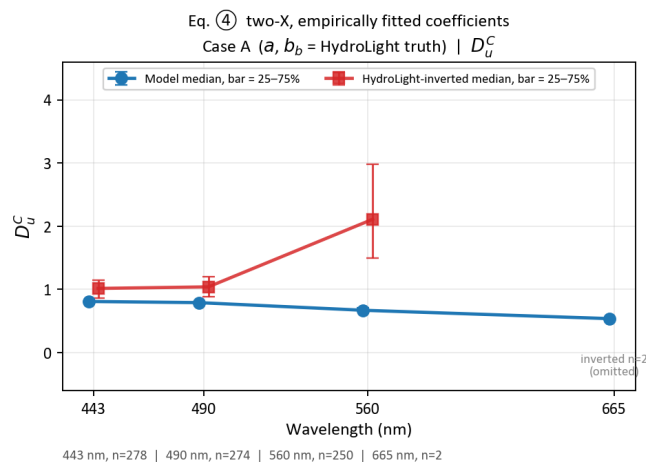


Figure A-7. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

Reading (Eq. ④ (two-X, empirical)). Empirical DuC (blue) is low, 0.81 (443 nm) → 0.53 (665 nm). Inverted medians (red) are 1.02 (443 nm), 1.04 (490 nm), 2.11 (560 nm): within 0.2 of blue at 443–490 nm and farther at 560 nm. 665 nm has only 2 successful inversions and is omitted (the bottom term is a large constant 1.613, leaving almost no room to fit the water-column term). In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

DuC (water-column term) — band-wise RMSE

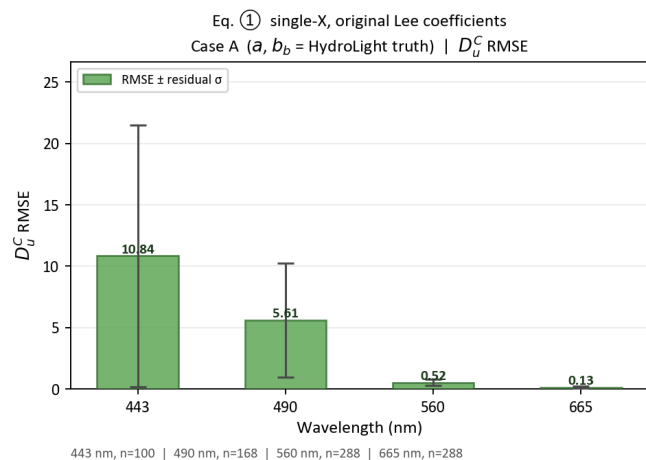


Figure A-9. Eq. ① single-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ① (single-X, Lee)). Case A RMSE from 443 to 665 nm: 10.84, 5.61, 0.52, 0.13; Case B: 9.23, 5.45, 0.73, 0.35. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

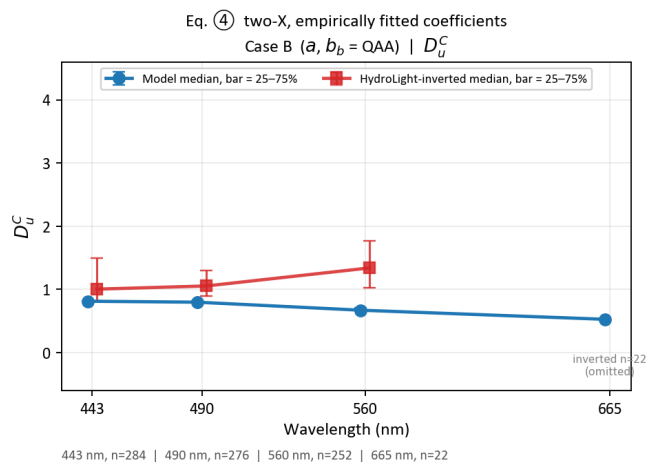


Figure A-8. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): DuC spectrum. Wavelength (nm) on the x-axis; DuC (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; n < 30 is marked omitted.

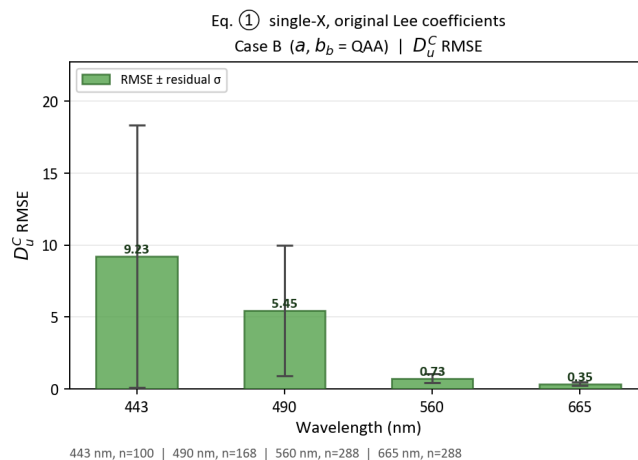


Figure A-10. Eq. ① single-X, original Lee coefficients, Case B (a, b_b = QAA): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

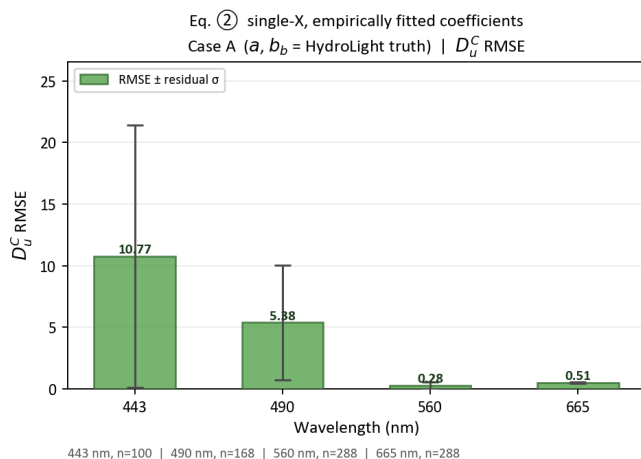


Figure A-11. Eq. ② single-X, empirically fitted coefficients, Case A ($a, b_b = \text{HydroLight truth}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ② (single-X, empirical)). Case A RMSE from 443 to 665 nm: 10.77, 5.38, 0.28, 0.51; Case B: 9.17, 5.23, 0.35, 0.29. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

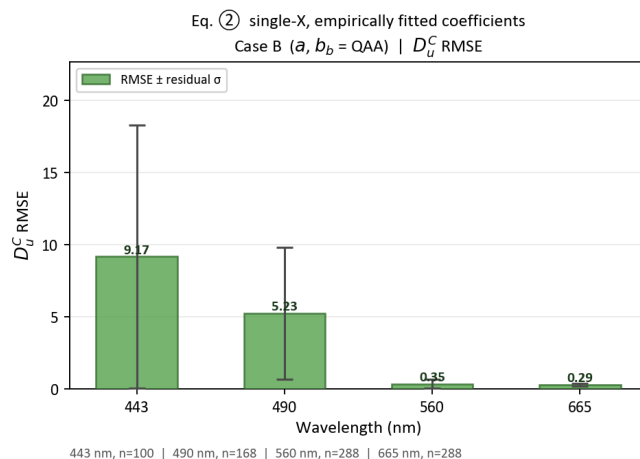


Figure A-12. Eq. ② single-X, empirically fitted coefficients, Case B ($a, b_b = \text{QAA}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

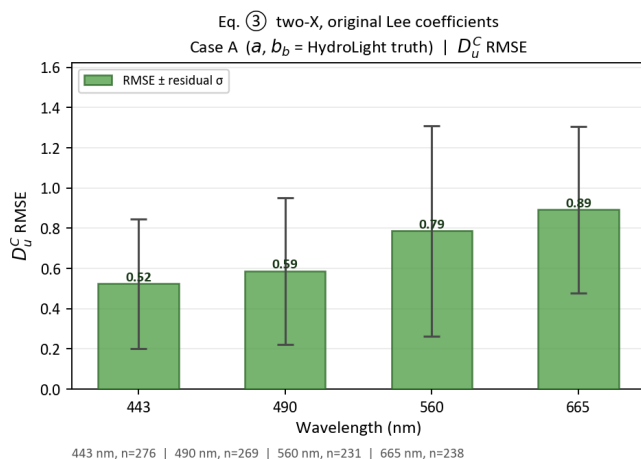


Figure A-13. Eq. ③ two-X, original Lee coefficients, Case A ($a, b_b = \text{HydroLight truth}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ③ (two-X, Lee)). Case A RMSE from 443 to 665 nm: 0.52, 0.59, 0.79, 0.89; Case B: 0.67, 0.51, 1.11, 1.47. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

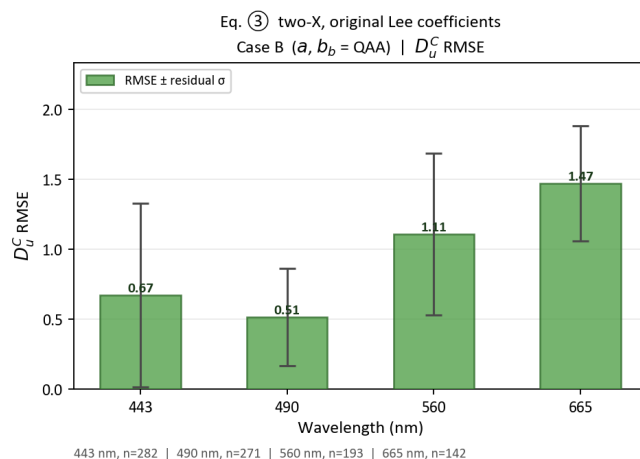


Figure A-14. Eq. ③ two-X, original Lee coefficients, Case B ($a, b_b = \text{QAA}$): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm \sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

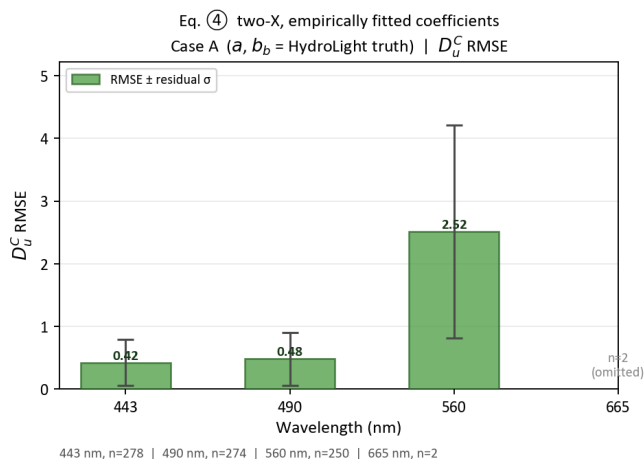


Figure A-15. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ④ (two-X, empirical)). Case A RMSE from 443 to 665 nm: 0.42, 0.48, 2.52, —; Case B: 1.06, 0.63, 1.28, —. Larger values mean the formula Du is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

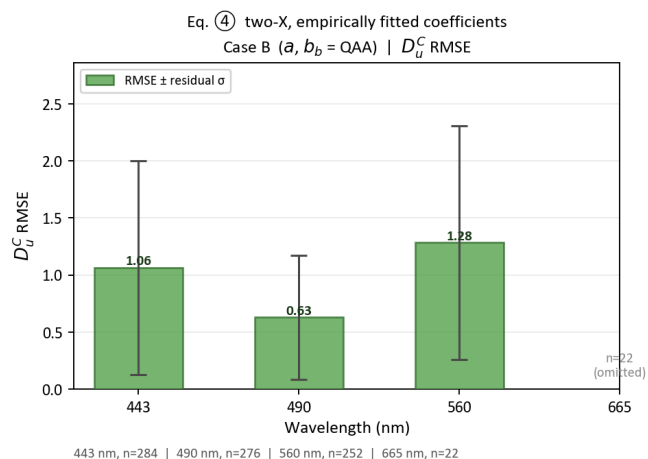


Figure A-16. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): band-wise DuC RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Eq.	Case A (HydroLight a, bb)				Case B (QAA a, bb)			
RMSE	443	490	560	665	443	490	560	665
①	10.84	5.61	0.52	0.13	9.23	5.45	0.73	0.35
②	10.77	5.38	0.28	0.51	9.17	5.23	0.35	0.29
③	0.52	0.59	0.79	0.89	0.67	0.51	1.11	1.47
④	0.42	0.48	2.52	— (n=2)	1.06	0.63	1.28	— (n=22)

Table A-1. Band-wise DuC RMSE (formula – HydroLight-inverted; dimensionless). Matches the bar plots above. Orange background = Case B.

DuB (bottom term; two-X Eqs. ③ and ④ only) — spectra

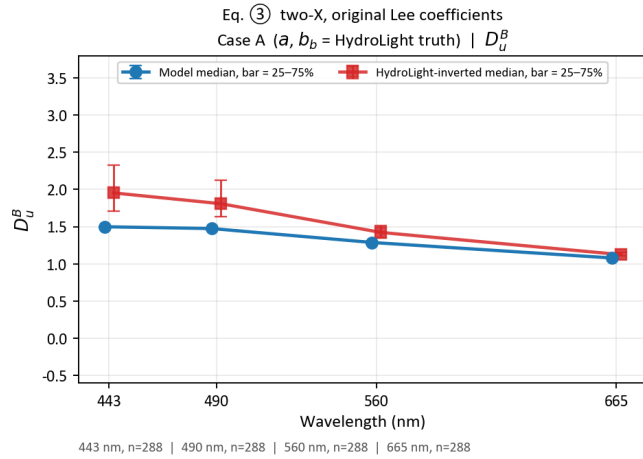


Figure A-17. Eq. ③ two-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

Reading (Eq. ③ (two-X, Lee)). Bottom term DuB: original Lee $1.04\sqrt{1+5.4u}$ (blue) is 1.50 (443 nm) $\rightarrow$ 1.08 (665 nm); inverted medians (red) are 1.95 (443 nm) $\rightarrow$ 1.12 (665 nm). At short wavelengths HydroLight requires stronger bottom attenuation than Lee. $n = 288$ at all four bands. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

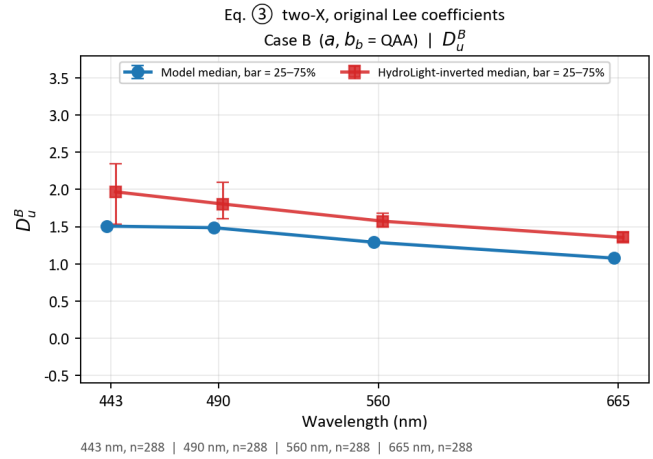


Figure A-18. Eq. ③ two-X, original Lee coefficients, Case B (a, b_b = QAA): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

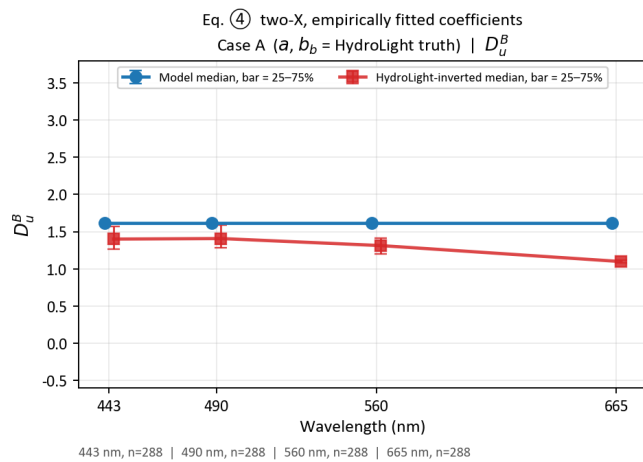


Figure A-19. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

Reading (Eq. ④ (two-X, empirical)). Empirical DuB = 1.613 constant (blue). Inverted medians (red) are 1.40 (443 nm) $\rightarrow$ 1.10 (665 nm); blue is 0.2–0.5 larger. A slightly large bottom term and a small water-column term is the combination that matches Rrs. In Case B (right), a and b_b come from QAA, so κ and u change and both the location and the spread of the inverted values change with them.

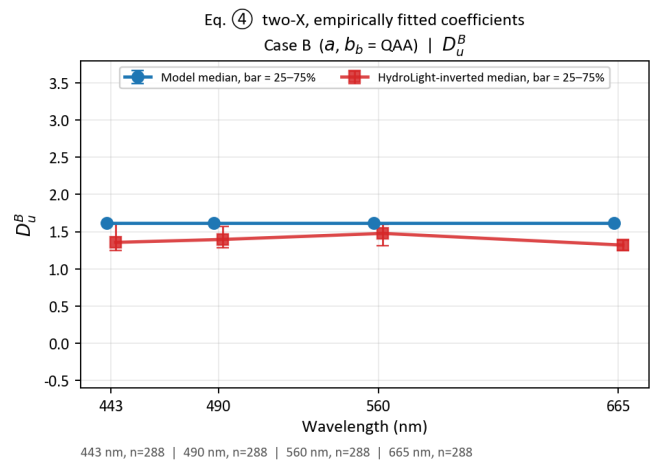


Figure A-20. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): DuB spectrum. Wavelength (nm) on the x-axis; DuB (dimensionless) on the y-axis. Blue circles = formula; red squares = HydroLight-inverted value. Points = median; vertical bars = 25–75% interval (medians are used because inverted values have a long tail). n under the panel = successful inversions; $n < 30$ is marked omitted.

DuB (bottom term; two-X Eqs. ③ and ④ only) — band-wise RMSE

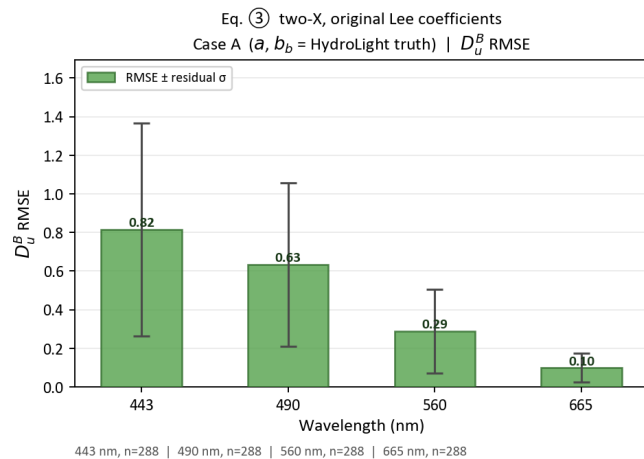


Figure A-21. Eq. ③ two-X, original Lee coefficients, Case A (a, b_b = HydroLight truth): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ③ (two-X, Lee)). Case A RMSE from 443 to 665 nm: 0.82, 0.63, 0.29, 0.10; Case B: 1.04, 0.63, 0.43, 0.31. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

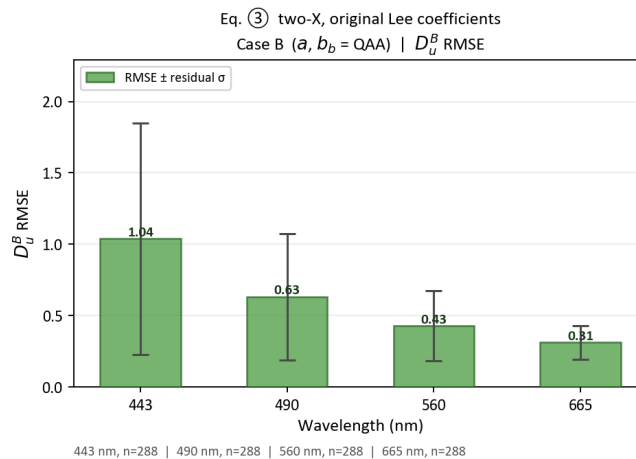


Figure A-22. Eq. ③ two-X, original Lee coefficients, Case B (a, b_b = QAA): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

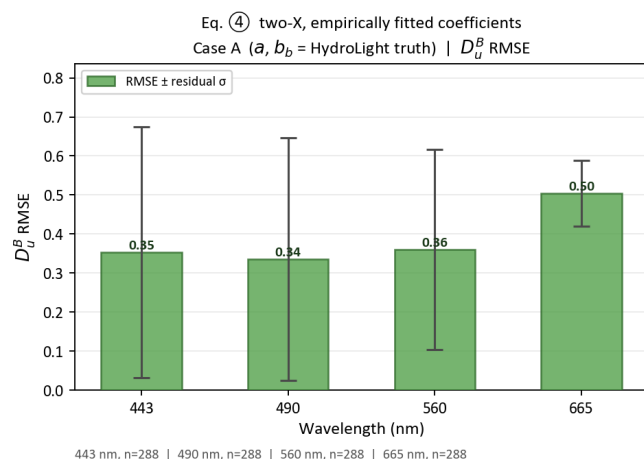


Figure A-23. Eq. ④ two-X, empirically fitted coefficients, Case A (a, b_b = HydroLight truth): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Reading (Eq. ④ (two-X, empirical)). Case A RMSE from 443 to 665 nm: 0.35, 0.34, 0.36, 0.50; Case B: 0.37, 0.33, 0.31, 0.31. Larger values mean the formula D_u is farther from the HydroLight requirement at that band. Bands with fewer than 30 successful inversions are omitted from the figures and tables.

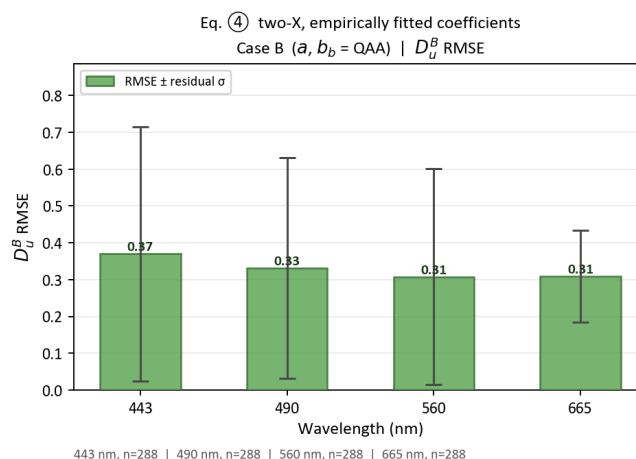


Figure A-24. Eq. ④ two-X, empirically fitted coefficients, Case B (a, b_b = QAA): band-wise DuB RMSE (formula – HydroLight-inverted). Wavelength (nm) on the x-axis; RMSE (dimensionless) on the y-axis. Green bars = RMSE (numbers on bars); black whiskers = residual $\pm\sigma$. The y-axis range differs by panel. Bands with fewer than 30 successful inversions are marked 'omitted' instead of a bar.

Eq.	Case A (HydroLight a, b_b)				Case B (QAA a, b_b)			
RMSE	443	490	560	665	443	490	560	665
③	0.82	0.63	0.29	0.10	1.04	0.63	0.43	0.31
④	0.35	0.34	0.36	0.50	0.37	0.33	0.31	0.31

Table A-2. Band-wise DuB RMSE (formula – HydroLight-inverted; dimensionless). Matches the bar plots above. Orange background = Case B.

Item 3 — Bottom endmembers: 6 types to 4 types

Four groups by Euclidean distance, one representative per group | 2026-09-03

Problem

The inversion uses four bands (443, 490, 560, 665 nm). Using six bottom types at once leaves too many unknowns. Coral, brown algae, red algae, and green algae are all dark, so they are hard to tell apart. The comment was to start with only 3–4 types that differ clearly in brightness.

Aim

Merge the six groups already defined in step 4 into four groups using Euclidean distance, and pick one representative in each group. These four types become the endmembers for the mixed-bottom inversion.

Method

Read reflectance at 443, 490, 560, and 665 nm. From these four values form a 7-D feature vector f , and measure similarity with Euclidean distance $d(x, y)$ (equations below).
To form four groups, compute d among the six group-mean feature vectors ($n = 7$; Table 1, Figure 1) and merge nearby groups. To pick a representative, compute the 4-band mean (centroid) of the members and take the original six-group representative with the smallest 4-band distance ($n = 4$) to that centroid. This keeps the endmembers consistent with the HydroLight runs already used.

$$f = [\bar{R}, R_{490}/R_{443}, R_{560}/R_{490}, R_{665}/R_{560}, R_{490} - R_{443}, R_{560} - R_{490}, R_{665} - R_{560}]$$

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$\bar{R}$ is the 4-band mean reflectance. Use $n = 7$ when merging groups and $n = 4$ when choosing a representative.

Distances among the six groups are given in Table 1 and Figure 1. The closest pair is G3–G4 (coral–dark vegetation, 0.29), then G1–G2 (sand–sediment, 0.36). Green algae (G6) is at least 1.58 from every other group. From this, dark vegetation and coral (G3+G4+G5) were merged, green algae was kept alone, and bright sand and sediment were kept as two groups because they occupy different brightness levels. That yields four groups.

Table 1. Euclidean distance between the six group-mean feature vectors (smaller = more similar)

	G1 Bright sand	G2 Medium sediment	G3 Coral / turf	G4 Dark vegetation	G5 Red algae	G6 Green algae
G1 Bright sand	—					
G2 Medium sediment	0.36	—				
G3 Coral / turf	0.93	0.61	—			
G4 Dark vegetation	1.10	0.77	0.29	—		
G5 Red algae	0.95	1.11	1.51	1.65	—	
G6 Green algae	2.50	2.23	1.65	1.58	2.93	—

Note: Yellow cells are the two closest pairs (G3–G4, G1–G2). The distance is $d(x, y)$ above with $n = 7$.

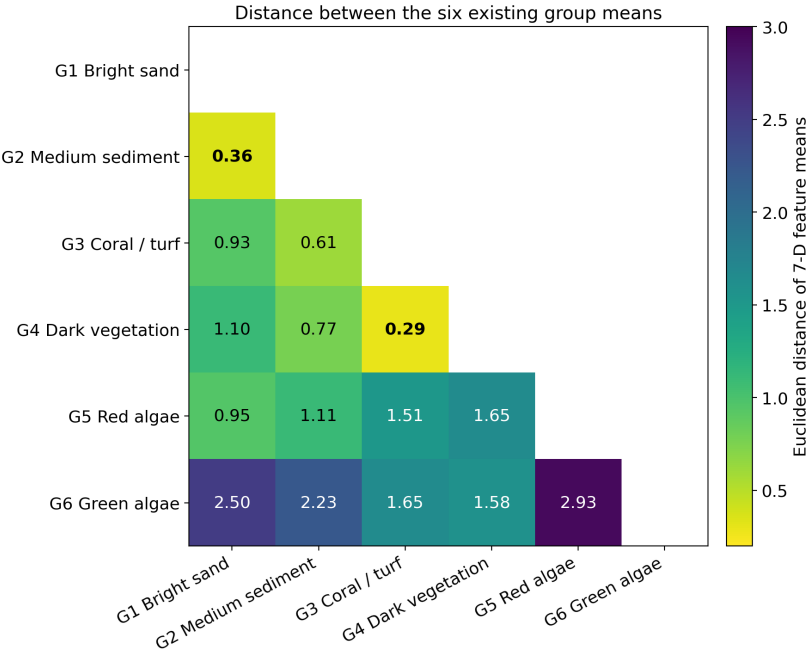


Figure 1. Same distances as Table 1. Yellow is close, dark blue is far. G3–G4 (0.29) and G1–G2 (0.36) are close; green algae (G6) is far from all groups.

Results

The four groups and representatives are listed in Table 2. The four representatives are ooid sand, dark sediment, brown algae, and green algae. Their 4-band mean reflectances are 0.55, 0.24, 0.04, and 0.12, so brightness is separated, and only green algae has a 560 nm peak (Figure 2). In the dark group, among the original six representatives, brown algae (0.059) was closer to the centroid than coral (0.078) or red algae (0.074).

Table 2. Four groups and representatives (distance = 4-band Euclidean distance)

Group	Members merged	Representative	Distance to centroid
Bright sand	ooid sand, coral sand	ooid sand	0.117
Medium sediment	hardpan, dark sediment	dark sediment	0.088
Dark vegetation / coral	coral, turf algae, kelp, brown algae, red algae, seagrass, etc.	brown algae	0.059
Green algae	green algae	green algae	0.000

Note: For two-member groups the two distances are equal, so the existing representative is kept.

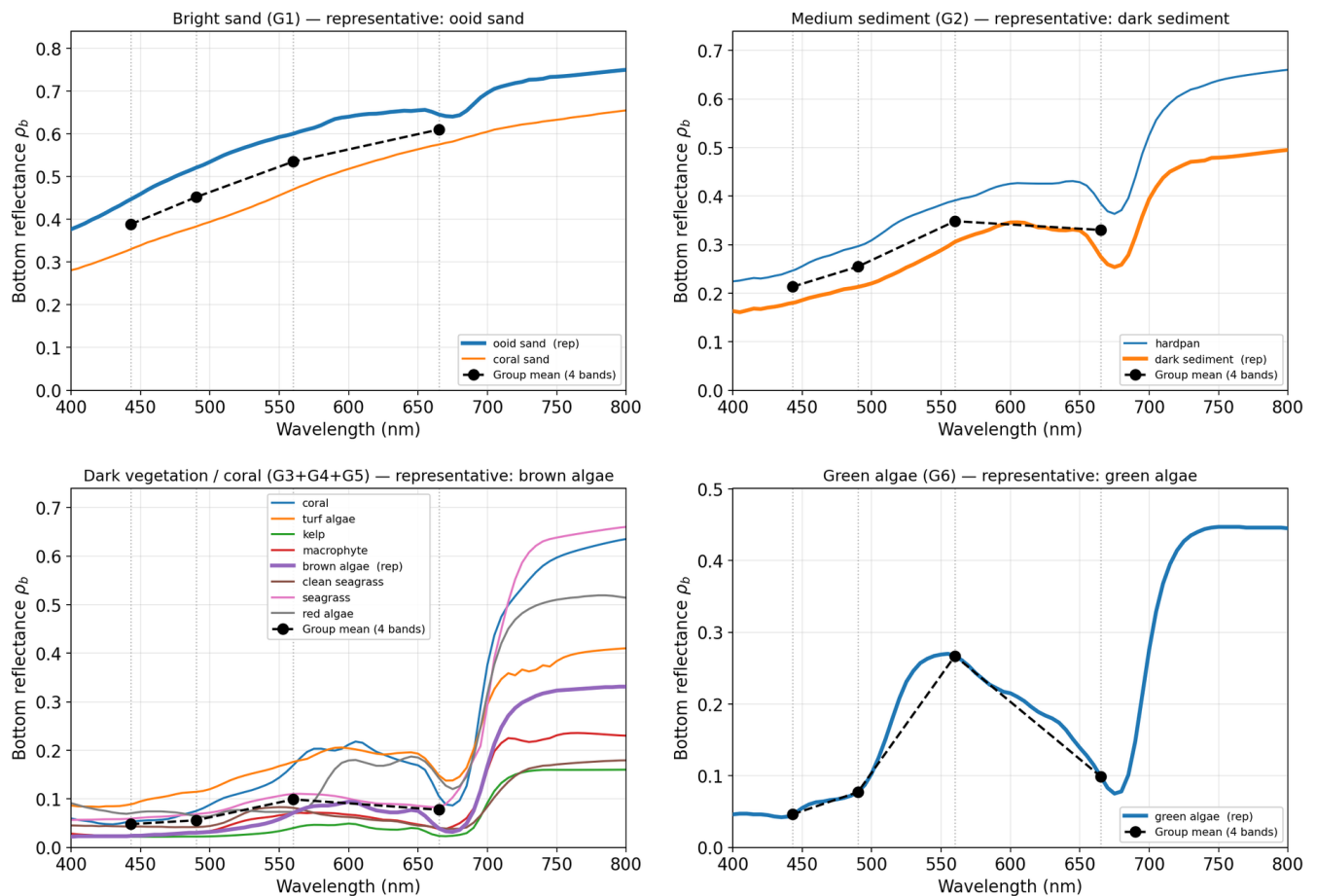


Figure 2. Spectra of the four groups. X-axis: wavelength (nm); Y-axis: bottom reflectance ρ_b . Thick line = representative; thin lines = other members; black dashed line with dots = 4-band group mean. Top left: bright sand; top right: sediment; bottom left: dark vegetation/coral; bottom right: green algae.

Conclusion

Conclusion

The mixed-bottom inversion uses four endmembers: ooid sand, dark sediment, brown algae, and green algae. Coral, red algae, and the rest will be added later.

Item 4 briefing — Selection of the depth-retrieval criterion

Comparison of three depth-retrieval methods | HydroLight independent set of 288 cases | 2026-09-03

Problem

The existing algorithm estimates a single common depth z from the R_{rs} reconstruction error over four bands (443, 490, 560 and 665 nm) by scalar minimization of a cost function. That multi-band procedure was not stated clearly enough in earlier reports, so the algorithm could be read as if a different depth were obtained at each wavelength, with no stated rule for choosing a final depth. It was therefore proposed that a depth $H(\lambda_k)$ be computed independently in each band, that R_{rs} be reconstructed from each of those depths, and that candidate solutions be ranked by the mean reconstruction residual.

Objective

The aim was to distinguish the existing four-band scalar-minimization scheme from the proposed per-band depth and mean-residual scheme, and to compare their depth-retrieval accuracy so that a more stable inversion criterion could be selected. The comparison also included the use of all four bands versus the three bands of relatively higher reliability for depth retrieval.

Procedure

The same validation set was inverted with three depth-retrieval methods. The difference is only what is minimized, as written in Table 1. Method 1 computes a depth in each band and ranks candidates by the mean residual ϵ^- ; Method 2 finds one common depth z^* that minimizes the sum of squared errors over four bands; Method 3 uses the same cost as Method 2 but on three bands only.

The validation set comprises 288 HydroLight cases (6 bottom types $\times$ 6 depths $\times$ 2 solar zenith angles $\times$ 2 chlorophyll levels $\times$ 2 suspended-matter levels). These 288 cases do not overlap the 972 cases used to fit the coefficients of Eqs. 2 and 4. All three methods were given the true bottom reflectance; absorption a and backscattering b_b were taken from QAA. Retrieved depths were compared with true depths using RMSE and MAE, separately for Eqs. 1–4.

Table 1. The three methods — only the quantity being minimized differs

Method	In plain words	Equation
Method 1 (proposed)	(1) At 443, 490 and 560 nm, find a depth H_k that matches that band alone. (2) Reconstruct all four bands with each H_k and take the mean residual ϵ_k^- . (3) Use the H_k whose ϵ_k^- is smallest.	$H_k = H(\lambda_k), k = 1, \dots, N$ $\epsilon_k^- = (1/M) \sum_j R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j; H_k) $ $\text{Cost} = \epsilon^- = (1/N) \sum_k \epsilon_k^-$
Method 2 (existing 4-band)	Place a single depth in the forward model, sum the squared errors over four bands, and take the common depth z^* that makes that sum smallest. No per-band depth is chosen.	$z^* = \arg \min_z \sum_{\lambda} [R_{rs}^{input}(\lambda) - R_{rs}^{sim}(\lambda; z)]^2$ $\lambda = 443, 490, 560, 665 \text{ nm}$
Method 3 (reliable 3-band)	Same equation as Method 2, but the cost uses only 443, 490 and 560 nm.	$z^* = \arg \min_z \sum_{\lambda} [R_{rs}^{input}(\lambda) - R_{rs}^{sim}(\lambda; z)]^2$ $\lambda = 443, 490, 560 \text{ nm}$

Notation. N = number of trusted bands (3); M = number of bands compared (4). R_{rs}^{input} is the observed (simulated) reflectance; $R_{rs}^{sim}(\lambda; H)$ is the value reconstructed at depth H . The Method-1 cost ϵ^- scores a candidate; the representative depth is the H_k with the smallest ϵ_k^- .

Why three bands were treated as the reliable set

When the true depth is substituted into the forward model, as in Item 1, the relative R_{rs} error is 2–8% at 443, 490 and 560 nm, but 13–21% at 665 nm—more than three times larger (Figure 1). R_{rs} itself is small at 665 nm, so a given absolute mismatch appears large as a relative error. For that reason 665 nm was treated as the less reliable band, and Methods 1 and 3 used only 443, 490 and 560 nm.

Band-wise R_{rs} relative error at true depth | orange = 665 nm | n = 288 per band per equation

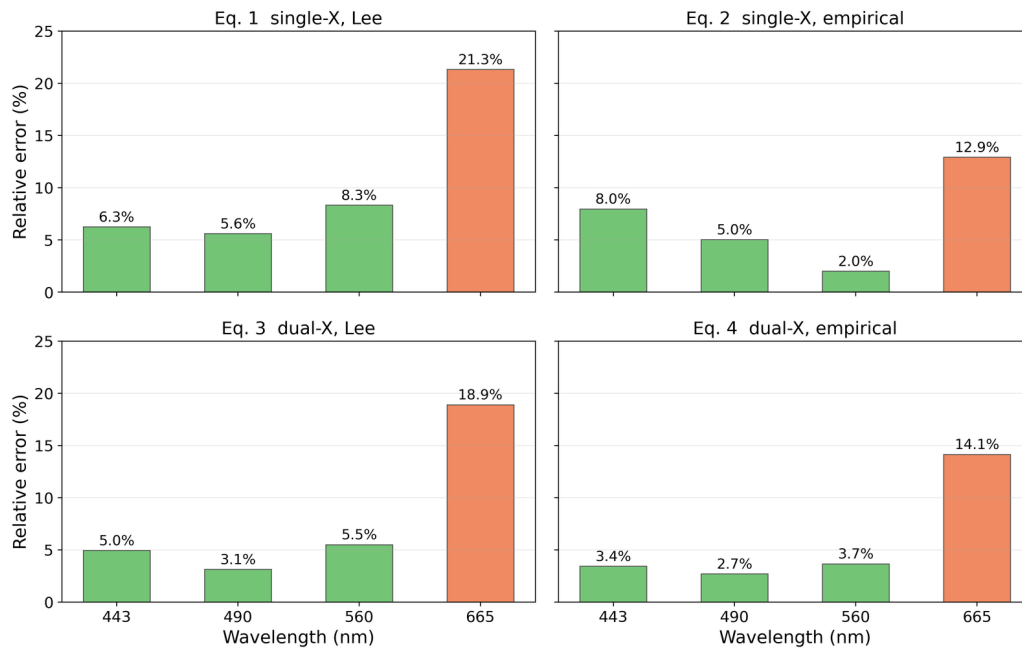


Figure 1. Band-wise relative R_{rs} error when the true depth is substituted (Eqs. 1–4; n = 288 per band). Green = 443, 490 and 560 nm; orange = 665 nm. Relative error at 665 nm alone is 13–21%, which is why Methods 1 and 3 omit that band.

Results

1) Depth accuracy — Method 2 is the most accurate

Depths for the 288 cases were retrieved by the three methods and compared with the true depths (Figure 2, Table 2). Method 2 was the most accurate (RMSE 0.39 m), followed by Method 3 (0.42 m) and Method 1 (0.50 m). Under Method 1 the three per-band depths differed by 1–2 m on average even for the same case, and depths from 443 and 490 nm were highly inaccurate. The band with the smallest residual was 560 nm in 62–72% of cases, so Method 1 effectively reduced to a 560 nm retrieval. Method 3, by dropping 665 nm, lost information that is sensitive to shallow depths and was therefore worse than Method 2.

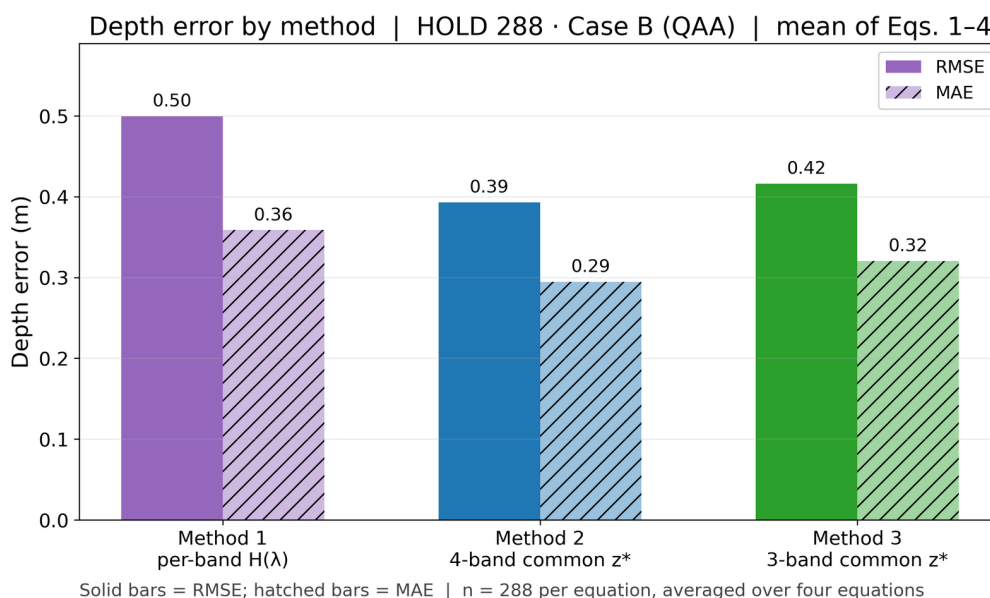


Figure 2. Depth error of the three methods (mean of Eqs. 1–4). Solid bars are RMSE and hatched bars are MAE; numbers above the bars are the plotted values. Method 2 (blue) is the smallest.

Table 2. Depth error by method (288 cases; mean of Eqs. 1–4)

Method	RMSE (m)	MAE (m)	MAPE (%)
Method 1 — selected per-band depth	0.50	0.36	15.4
Method 2 — four-band common depth	0.39	0.29	11.9
Method 3 — three-band common depth	0.42	0.32	14.8

Note. The highlighted row is the selected Method 2. The ranking was unchanged when HydroLight true a and b_b were used, and when the comparison was repeated on the 972-case set.

2) Method 2 depth retrieval (Eqs. 1–4)

Retrieved depths from Method 2 are compared with true depths in Figure 3. All four equations follow the 1:1 line over 0.5–5 m. Equation-wise errors are given in Figure 4 and Table 3. Eq. 3 was the most accurate (RMSE 0.33 m); Eqs. 4 (0.35 m) and 2 (0.36 m) were similar, and Eq. 1 was the poorest (0.53 m). Equations with the original Lee coefficients (1 and 3) tended to overestimate depth; those with empirical coefficients (2 and 4) tended to underestimate it.

Method 2 depth retrieval (4-band squared cost) | HOLD 288 · Case B (QAA)

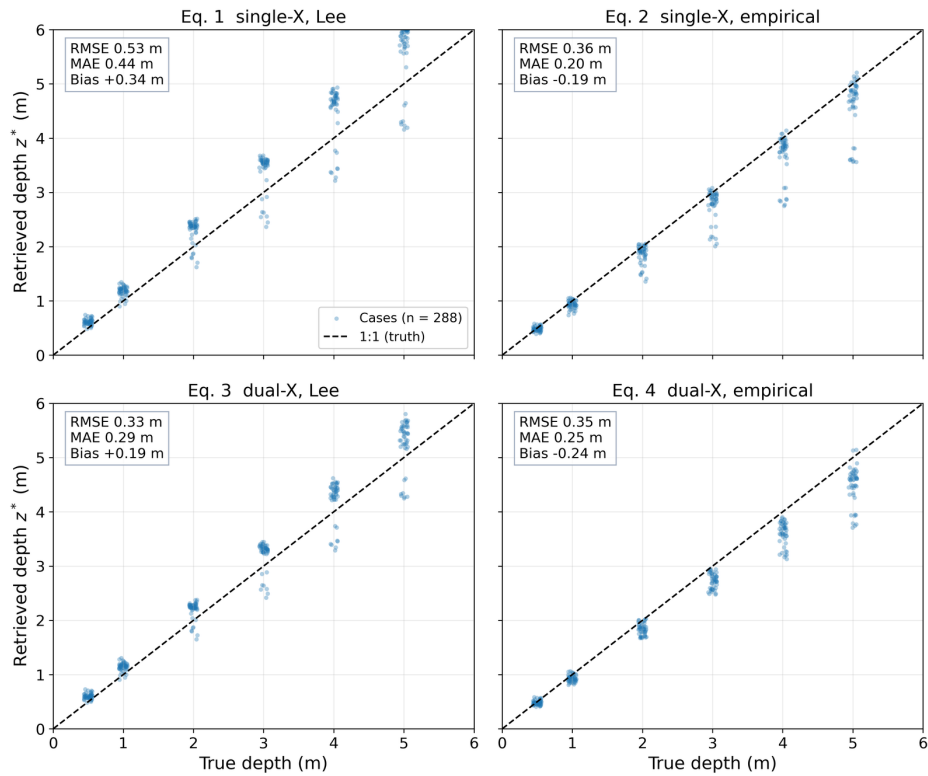
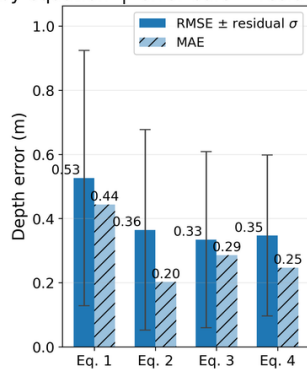


Figure 3. Method 2 depth retrieval (Eqs. 1–4; $n = 288$ each). The x-axis is true depth and the y-axis is retrieved depth; each point is one case. Closer to the dashed 1:1 line is more accurate. The inset box gives RMSE, MAE and bias for that equation.

Method 2 depth error by equation | error bars = residual σ | $n = 288$ per equation

Eq. 1 = single-X Lee; Eq. 2 = single-X empirical; Eq. 3 = dual-X Lee; Eq. 4 = dual-X empirical. Solid = RMSE; hatched = MAE; error bars = residual σ (mean-based, not median).

Figure 4. Method 2 depth error by equation ($n = 288$ each). Solid bars are RMSE and hatched bars are MAE; both are mean-based statistics (not medians). Vertical error bars on RMSE are the standard deviation of the residual (retrieved minus true). Eq. 3 is the smallest and Eq. 1 is the largest.

Table 3. Method 2 depth error by equation (288 cases per equation)

Equation	RMSE (m)	MAE (m)	Bias (m)	MAPE (%)
1 single-X, Lee coefficients	0.53	0.44	+0.34	18.1
2 single-X, empirical coefficients	0.36	0.20	-0.19	7.8
3 dual-X, Lee coefficients	0.33	0.29	+0.19	12.9
4 dual-X, empirical coefficients	0.35	0.25	-0.24	8.9

Note. Positive bias means the retrieval is too deep; negative bias means it is too shallow.

3) R_{rs} reconstructed from the retrieved depth (Eqs. 1–4)

The retrieved depth z^* was substituted back into the forward equation and the resulting R_{rs} was compared with HydroLight. In Figure 5 the model (blue) and HydroLight (red) means and medians nearly overlap for all four equations, showing that the spectrum is well reproduced at the retrieved depth. Band-wise RMSE (Figure 6) is larger at 560 nm, where the signal is strongest, and at 443 nm, where QAA error is larger, and is smallest at 490 nm. The error at 665 nm is comparable to that at 490 nm, so including 665 nm in the cost is not a problem.

R_{rs} reconstructed from Method-2 z^* | blue = model, red = HydroLight | $n = 288$ per equation

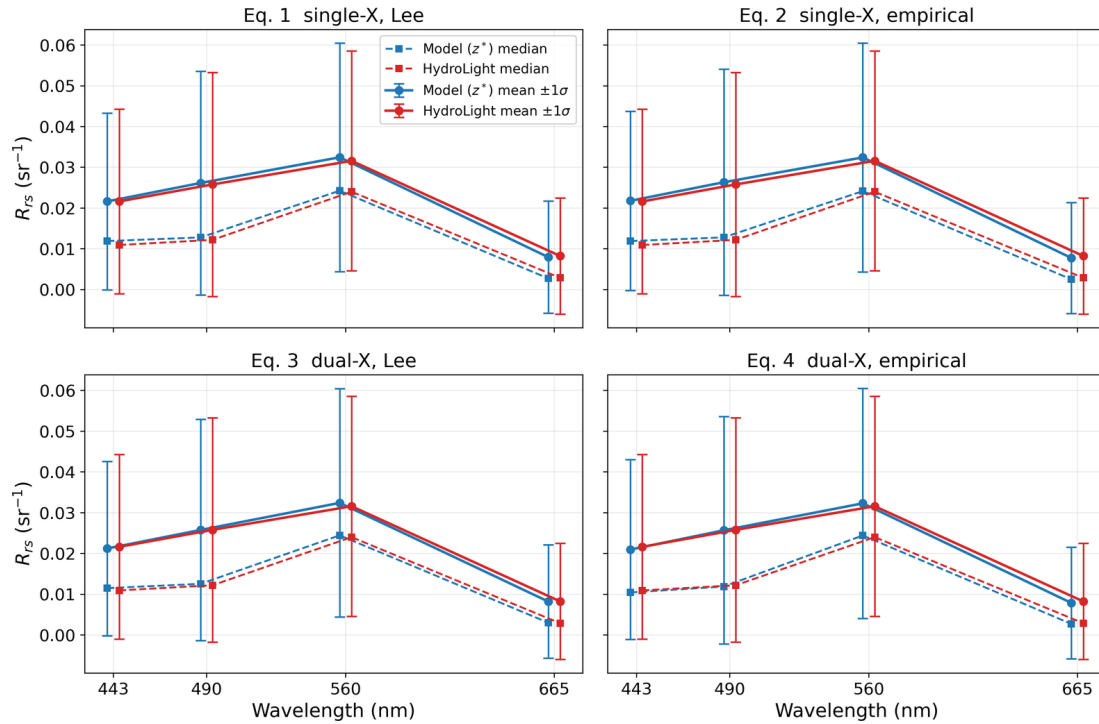


Figure 5. R_{rs} reconstructed from Method-2 depths (Eqs. 1–4; $n = 288$ each). Blue = model; red = HydroLight. Solid lines are means with vertical bars of $\pm 1\sigma$; dashed lines are medians. Closer overlap of the blue and red lines indicates better reconstruction.

Method 2 band-wise reconstructed R_{rs} RMSE | numbers = $\text{RMSE} (\times 10^{-3})$ | $n = 288$ per band per equation

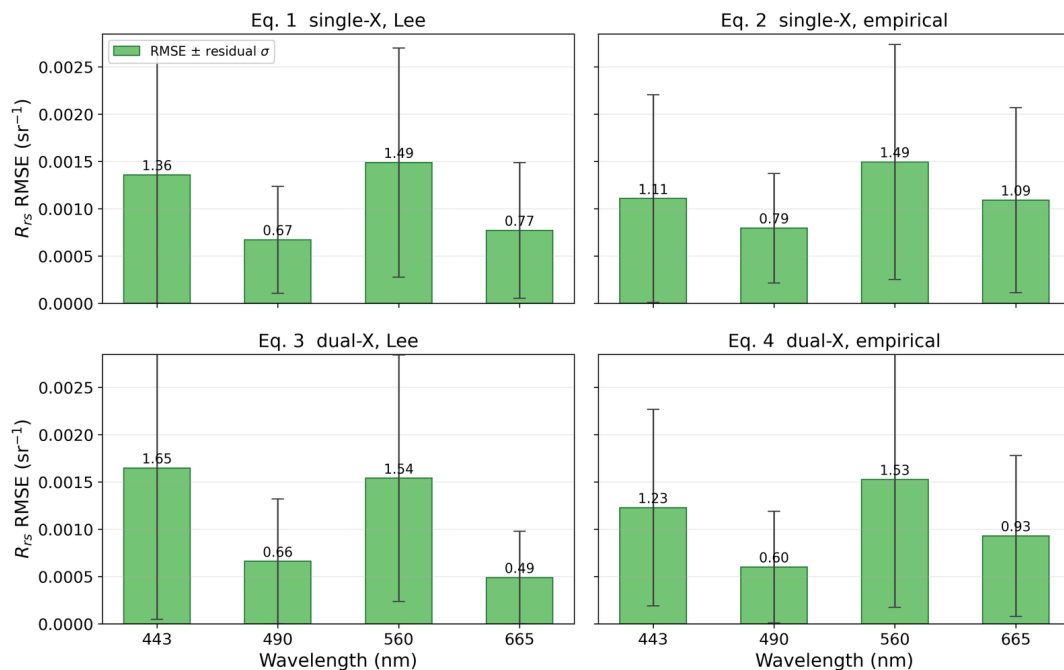


Figure 6. Method 2 band-wise R_{rs} RMSE (Eqs. 1–4; $n = 288$ per band). Bars are RMSE; numbers above the bars are $\text{RMSE} (\times 10^{-3})$; vertical error bars are the standard deviation of the residual.

Conclusion

Conclusion

When depth is obtained separately in each band, the three depths for the same case differ by 1–2 m on average and by as much as 3–5 m. Depths from 443 and 490 nm are particularly inaccurate, so Method 1 effectively selects the 560 nm depth and is less accurate than Method 2 (RMSE 0.50 m versus 0.39 m). Choosing one band as a representative depth is therefore not justified.

The depth-retrieval criterion is therefore Method 2: a single common depth z^* obtained by scalar minimization of a reconstruction-error cost over four bands (443, 490, 560 and 665 nm). There is no step that selects a band, and both the depth error and the R_{rs} reconstruction error are the smallest of the three methods. Relative error at 665 nm is large, but that band still carries information on shallow-depth change; omitting it (Method 3) degrades accuracy, so all four bands are retained. Eqs. 3 and 4 are used, with empirical-coefficient Eq. 4 preferred. These rankings were obtained on the 288 cases that were not used to fit coefficients, and they were unchanged with true a and b_b and on the 972-case set.

The proposed mean of per-band reconstruction residuals is not used to choose the depth. It is used in the next step to rank seafloor candidates: Method 2 supplies z^* for each candidate, and the mean reconstruction residual is then compared among candidates.

Limitation. This comparison used the true bottom reflectance. A comparison in which bottom type and mixing fractions are also unknown is left to Item 5 (mixing-fraction inversion).

Item 5 briefing — Joint retrieval of bottom mixing fractions and depth

Advisor procedure + Item-4 squared Cost | HydroLight validation data | 2026-09-03

Problem

The existing method treated the bottom as a single type and retrieved depth only. Real bottoms are mixtures of several materials, so not only depth but also how much of each material is present (the mixing fractions) must be matched. The comment was not to scan all mixing fractions on a 5–10% grid, but to take the top K by the residual of representative Seeds and then move the mixing fractions and depth together.

Objective

This note records the method that retrieves bottom mixing fractions and depth together. It distinguishes the original method (bottom fixed, depth only) from the method used by the advisor (mixing fractions and depth together). The three-band option was kept as a reference because, in Item 1, the relative error at 665 nm was large; it is not used in the Cost of this calculation (Item 4 showed that dropping 665 nm worsens depth). Results are from 6,250 cases run with the advisor procedure, comparing Cost as-is with Cost multiplied by 1 million, looking at both bottom fractions and depth.

What is used in this calculation

Cost = the sum of squared differences between observed R_{rs} and modelled R_{rs} . This was fixed in Item 4. The advisor's original formula (mean of absolute values) is not used.

The tolerance 0.000001 (10^{-6}) is the SLSQP default. The number itself was not changed. Only when the value is passed to the optimizer is Cost multiplied by 1 million (10^6). The 1 million is the reciprocal of the square of the leftover R_{rs} error (about 0.001) when the true answer is inserted in Item 1.

Validation data

The validation data are mixed-bottom R_{rs} generated with HydroLight. The true bottom to be matched is brown algae 80% + ooid sand 20%. Dark sediment and green algae are 0. The simulation conditions on the presentation slides are given in Table 1. The slides use 11 depth steps and 1,375 cases. This calculation uses the same bottom, water quality and solar zenith angles, but depth only at 0.1 m spacing (50 steps), giving 6,250 cases. The mixed HydroLight set is this single 80:20 grid. Other fractions and pure bottoms are not demonstrated by these numbers.

Table 1. HydroLight validation simulation conditions (presentation slides)

Item	Simulation conditions
Simulation	HydroLight radiative-transfer simulation
Water type	Classic Case-1 (Case-1 Water)
Bottom material (true)	Brown Algae 80% + Ooid Sand 20%
Depth	0.1, 0.5, 1.0, 1.5, ..., 4.5, 5.0 m (11 values) → this calculation: 0.1–5.0 m, 0.1 m spacing (50 values)
Solar zenith angle	10, 20, 30, 40, 50° (5 values)
Wavelength	443, 490, 560, 665 nm (4 values)
Chl-a concentration	0.1, 0.2, 0.3, 0.4, 0.5 mg/m ³ (5 values)
TSM concentration	0.1, 0.2, 0.3, 0.4, 0.5 mg/m ³ (5 values)
Water temperature	20.0 °C
Salinity	35 PSU
Sky / cloud	Clear sky, cloud fraction 0.0
Number of cases	Slides 1,375 cases (11×5×5×5). This calculation 6,250 cases (50×5×5×5)

Procedure

1) Original method — the bottom is already known; only depth is retrieved

The original method assumes the bottom type (reflectance) is known and retrieves depth only. Bottom mixing fractions are not retrieved. Cost is the sum of squared differences between observed R_{rs} and modelled R_{rs} . Because a pixel has one depth, the depth that minimizes this Cost over the four bands 443·490·560·665 nm is found. A depth is not obtained separately in each band and then one of them chosen.

Original method — Cost = sum of squared Rrs errors. Bottom fixed; depth only

$$Cost(z) = \sum_{\lambda} [R_{rs}^{obs}(\lambda) - R_{rs}^{model}(\lambda; z)]^2$$

$$\lambda = 443, 490, 560, 665 \text{ nm}$$

$$z^* = \arg \min_{z > 0} Cost(z)$$

Cost = sum of squared differences between observed Rrs and modelled Rrs

Figure 1. Cost = sum of squared R_{rs} errors. The bottom is fixed, and only the depth z^* that minimizes this value is retrieved. Mixing fractions are not obtained.

Table 2. Symbols in Figure 1

Symbol	Meaning
Cost	Sum of squared differences between observed R_{rs} and modelled R_{rs} . The error in each band is squared and added.
z	Depth to be retrieved (m). One per pixel.
R_{rs}^{obs}	HydroLight (or observed) remote-sensing reflectance
R_{rs}^{model}	Remote-sensing reflectance computed from the shallow-water formula at depth z
λ	Wavelength. The original method uses four: 443, 490, 560, 665 nm
z^*	Depth at which Cost (sum of squared R_{rs} errors) is smallest. Depth is greater than 0.

2) Original method using only three wavelengths

When true depth and true bottom are inserted and R_{rs} is computed, as in Item 1, the relative error at 443·490·560 nm is small, but at 665 nm alone it is 13–25% (Figure 2). R_{rs} itself is small at 665 nm, so the same difference appears large as a relative error. A variant that drops 665 nm from the original Cost and uses only three bands was also kept. In that case too, only depth is retrieved; bottom mixing fractions are not. However, Item 4 showed that dropping 665 nm worsens depth, so the Cost of this calculation keeps four bands. The three-band Cost is a reference and was not used on these 6,250 cases.

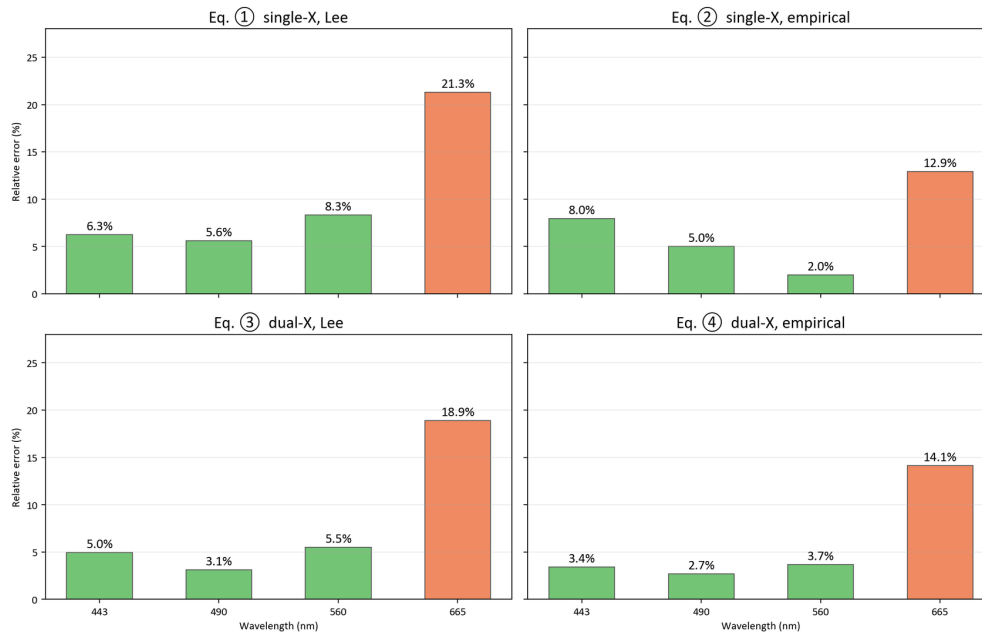
Band-wise R_{rs} relative error at true depth | orange = 665 nm | HOLD 288 per band per equation

Figure 2. Item 1 validation. Band-wise relative R_{rs} error when the true depth is substituted (HOLD 288). Green = 443-490-560 nm; orange = 665 nm. 665 nm alone is distinctly larger. The three-band Cost is the reference that omits it; the Cost of this calculation keeps four bands.

Three wavelengths only — Cost = sum of squared R_{rs} errors, 665 nm omitted

$$Cost(z) = \sum_{\lambda} [R_{rs}^{obs}(\lambda) - R_{rs}^{model}(\lambda; z)]^2$$

$\lambda = 443, 490, 560 \text{ nm}$ (665 nm excluded)

Figure 3. Three-band Cost = sum of squared R_{rs} errors. The formula is the same as the original method; only the wavelengths are 443-490-560 nm.

Table 3. How Figure 3 differs from the original method

Item	Original method	Three wavelengths only
Wavelengths	443, 490, 560, 665 nm	443, 490, 560 nm
Cost	Sum of squared R_{rs} errors over four bands	Sum of squared R_{rs} errors over three bands
What is retrieved	Depth only (bottom fixed)	Depth only (same)
Mixing fractions	Not retrieved	Not retrieved
This calculation	Cost is this four-band sum of squares	Reference only. Not used on these 6,250 cases

3) Method used by the advisor — bottom mixing fractions and depth are retrieved together

The method used by the advisor does not treat the bottom as a single type. The mixing fractions f_i of the four materials and the depth H are retrieved together. Mixing fractions are not scanned on a full grid. Representative Seeds are made, the mean residual (absolute value) of each Seed is inspected, and only the top K with the smallest residual are taken as initial values; then mixing fractions and depth are moved together. The Cost written by the advisor is not a square; it is the mean of the absolute band errors (J in Figure 4). In this calculation the Seeds are 4 pure + six 50:50 pairs, 10 in total, and the top $K = 5$. Ten is a small number, enough to stand as representatives instead of a full grid, and $K = 5$ is half of that. Three is too few, and ten would leave no selection step.

Three constraints were imposed so that the answer makes sense. ① The four material fractions must sum to 100% (the bottom cannot be empty or 200%). ② A single material cannot be less than 0% or more than 100%. ③ Depth must be greater than 0. It cannot be negative or 0 m. The formula used by the advisor has no upper bound on depth.

This calculation keeps the above procedure (Seed → top K → move together) unchanged, and changes only Cost to the Item-4 sum of squares, and only the depth range to 0.001–6 m. In Item 4 the sum of squares matched depth better, and the depth range is the same as Item 4.

Advisor method — retrieve bottom mixing fractions f and depth H together

1–3. Bottom and initial depth of the representative Seeds

$$\rho_b^{(s)}(\lambda) = \sum_i f_i^{(s)} \rho_i(\lambda), \quad \sum_i f_i^{(s)} = 1$$

$$H_s^{(0)} = \text{DepthEquation}(R_{rs}, \rho_b^{(s)}, a, b_b)$$

4–5. Mean residual of each Seed → top K

$$\varepsilon_s = \frac{1}{M} \sum_{j=1}^M |R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j)|$$

$$S_{TopK} = \text{TopK}_s(\varepsilon_s)$$

6. Optimize mixing fractions and depth together from the top K (SLSQP)

$$\min_{f, H} J(f, H), \quad J = \frac{1}{M} \sum_{j=1}^M |R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j; f, H)|$$

$$\sum_i f_i = 1, \quad 0 \leq f_i \leq 1, \quad H > 0$$

Fractions must sum to 100%; each material 0–100%; depth must be greater than 0.

7–8. Adopt the solution with the smallest final residual among the K

$$\varepsilon_k^{final} = \frac{1}{M} \sum_{j=1}^M |R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j; f_k^*, H_k^*)|$$

$$k^* = \arg \min_k \varepsilon_k^{final}, \quad H^* = H_{k^*}^*, \quad f^* = f_{k^*}^*$$

Figure 4. Procedure used by the advisor. Bottom fractions are set from Seeds (1–3), the top K are selected by mean residual (4–5), mixing fractions and depth are optimized together with SLSQP (6), and the mixing fractions f^* and depth H^* of the solution with the smallest residual are taken (7–8). J in the figure is the advisor's mean of absolute values. The Cost of this calculation is the sum of squares (Table 5).

Table 4. Symbols in Figure 4

Symbol	Meaning
f_i	Mixing fraction of material i. i = ooid, dark sediment, brown algae, green algae
ρ_i	Bottom reflectance of material i
$\rho_b^{(s)}$	Mixed bottom reflectance of Seed s
$H_s^{(0)}$	Initial depth obtained by the original method with that Seed bottom fixed
R_{rs}^{input}	Input (HydroLight) remote-sensing reflectance
R_{rs}^{sim}	Remote-sensing reflectance reconstructed from the formula
ε_s, J	Mean of the absolute band errors. M is the number of wavelengths used.
M	Number of wavelengths. The advisor formula is a mean of absolute values. With three bands, M = 3.
K	Number of Seeds passed to continuous optimization. 5 in this calculation.
H^*, f^*	Solution with the smallest final residual among the K. f^* is the bottom mixing fractions; H^* is depth.
$\sum f_i = 1$	The four material fractions must sum to 100%. The bottom cannot be empty or 200%.
$0 \leq f_i \leq 1$	A single material cannot be less than 0% or more than 100%.
$H > 0$	Advisor original formula: depth must be greater than 0, no upper bound. This calculation: 0.001–6 m (Table 5).

Table 5. Settings used in this calculation

Item	Setting
Materials	ooid sand, dark sediment, brown algae, green algae (the four types fixed in Item 3)
True values to match	ooid 20%, brown 80%, dark 0%, green 0%
Seeds	4 pure + six 50:50 pairs, 10 in total. Representatives instead of a full grid.
Top K	The 5 with the smallest initial residual (half of 10. Three is too few; ten would leave no selection step.)
Cost	Sum of squares fixed in Item 4. The advisor's original formula (mean of absolute values) is not used.
Tolerance	0.000001 (10^{-6} , SLSQP default. The number is left as is.)
Value passed to SLSQP	Cost $\times$ 1 million (10^6). Reciprocal of the square of the Item-1 forward R_{rs} error of about 0.001.
Sum of fractions	The four materials must sum to 100%.
Each fraction	Cannot be less than 0% or greater than 100%.
Depth	0.001–6 m (same as Item 4. The advisor's original formula has no upper bound.)

Problem that arose in the procedure

Problem that arose in the procedure

The stopping criterion (tolerance) was not decided at the outset. 0.000001 (10^{-6}) was used at first because it is the SLSQP default. It was not a number chosen after looking at the size of Cost.

Cost is the sum of squared differences between observed R_{rs} and modelled R_{rs} . R_{rs} itself is small, so Cost is very small, 10^{-7} – 10^{-6} . It is already below the tolerance 0.000001 (10^{-6}), so the optimizer treats it as “already small enough” and stops at once. Depth and bottom mixing fractions are therefore decided quickly, barely moving from the Seed starting values. The mixing fractions are half dark and half brown, and ooid is 0.

The tolerance was not made larger. Making the tolerance larger would stop even more easily. The second calculation left the tolerance number at 0.000001 (10^{-6}) and multiplied only Cost (squared R_{rs} error) by 1 million (10^6) to match the scale of the numbers. The 1 million is the reciprocal of the square of the leftover R_{rs} error (about 0.001) when the true answer is inserted in Item 1. The location of the minimum (argmin) does not change. This is the stopping rule of this calculation.

Results

The tolerance is the SLSQP default 0.000001 (10^{-6}) in both runs. The only difference is whether Cost (sum of squared R_{rs} errors) was passed as-is or multiplied by 1 million (10^6). The bottom to be matched is ooid 20%, brown 80%.

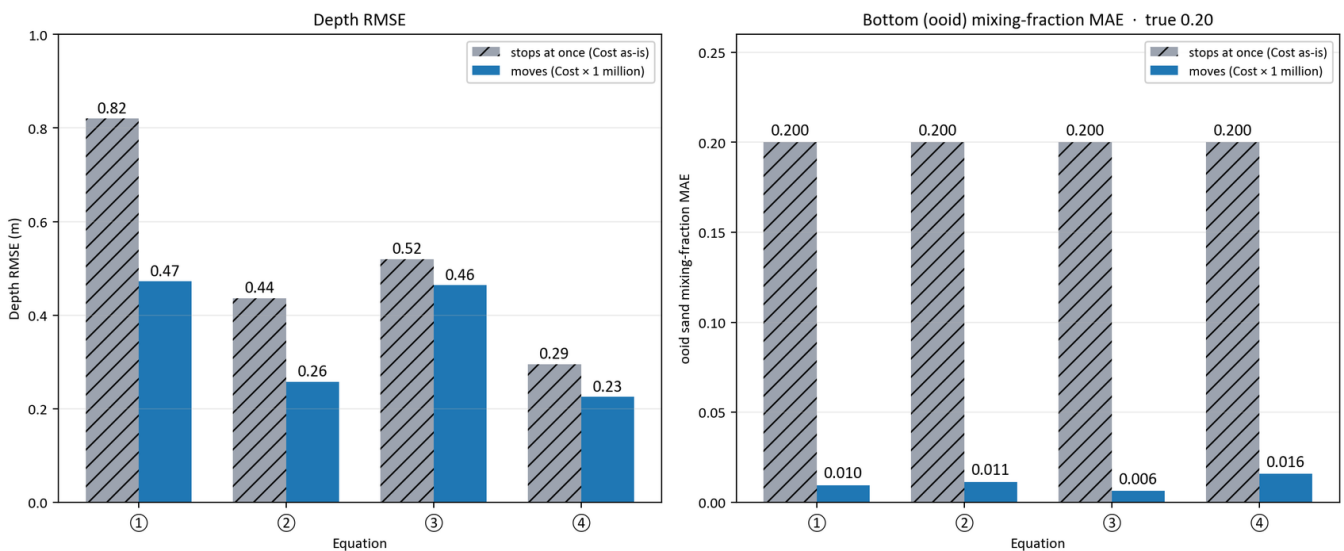


Figure 5. Hatched = Cost as-is (stops at once); blue = Cost × 1 million (moves). Tolerance 0.000001 (10^{-6}) is the SLSQP default. $n = 6250$ per equation.

Looking at Figure 5 alone, Eqs. ② and ④ are good. Depth RMSE is 0.23 m for ④ and 0.26 m for ②. ① and ③ are large, 0.46–0.47 m. The hatched bars on the right are all 0.200 because sand was left at 0. Blue found the sand (0.006–0.016).

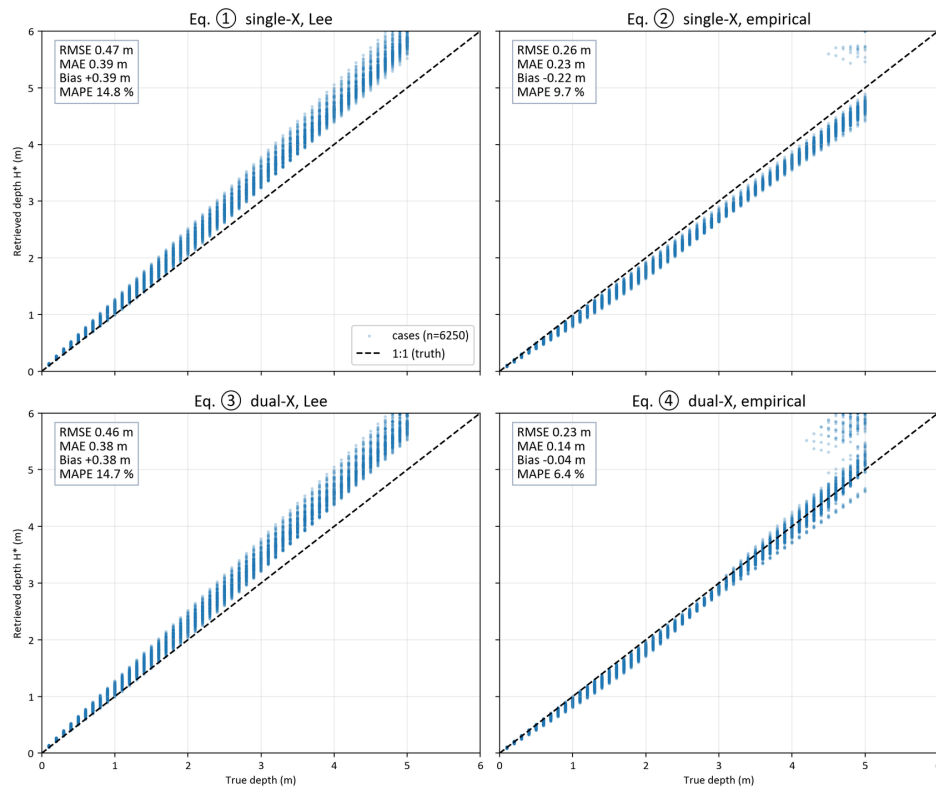
Depth from joint retrieval of mixing fractions and depth | brown 80% + ooid 20% · QAA a , b_b 

Figure 6. Depth scatter. x-axis = true depth; y-axis = retrieved depth. The black dashed line is 1:1 (truth). Closer to the line means depth is matched. $n = 6250$ per equation.

Figure 6 is the figure that shows whether depth is matched, as it is. For Eqs. ② and ④ the points lie near the 1:1 line. Eqs. ① and ③ rise well above the line and take depth too large. This is the same story as the left panel of Figure 5.

Table 6. After Cost $\times$ 1 million — depth, mixing fractions and reconstructed R_{rs} (QAA, $n = 6250$ per equation)

Equation	Depth RMSE (m)	Mean mixing fractions (ooid, dark, brown, green)	ooid MAE	brown MAE	Reconstructed R_{rs} RMSE	R_{rs} relative RMSE (%)
① single-X, Lee coefficients	0.47	(0.19, 0.00, 0.80, 0.01)	0.010	0.012	0.67×10^{-4}	1.11
② single-X, empirical coefficients	0.26	(0.19, 0.00, 0.81, 0.00)	0.011	0.013	0.95×10^{-4}	1.93
③ dual-X, Lee coefficients	0.46	(0.20, 0.01, 0.79, 0.00)	0.006	0.015	0.58×10^{-4}	0.94
④ dual-X, empirical coefficients	0.23	(0.20, 0.02, 0.77, 0.00)	0.016	0.027	1.90×10^{-4}	4.03

Note. True mixing fractions (0.20, 0, 0.80, 0). Blue rows = Eqs. ② and ④, which have good depth in Figure 5. Reconstructed R_{rs} RMSE is the value obtained by running the formula again with the retrieved (mixing fractions, depth) and comparing with HydroLight.

On depth alone, ④ is first. Looking at depth and bottom together, ② is acceptable on both. Reconstructed R_{rs} is smallest for ③ (0.58×10^{-4}) and largest for ④ (1.90×10^{-4}). A good depth does not mean R_{rs} is the best match as well.

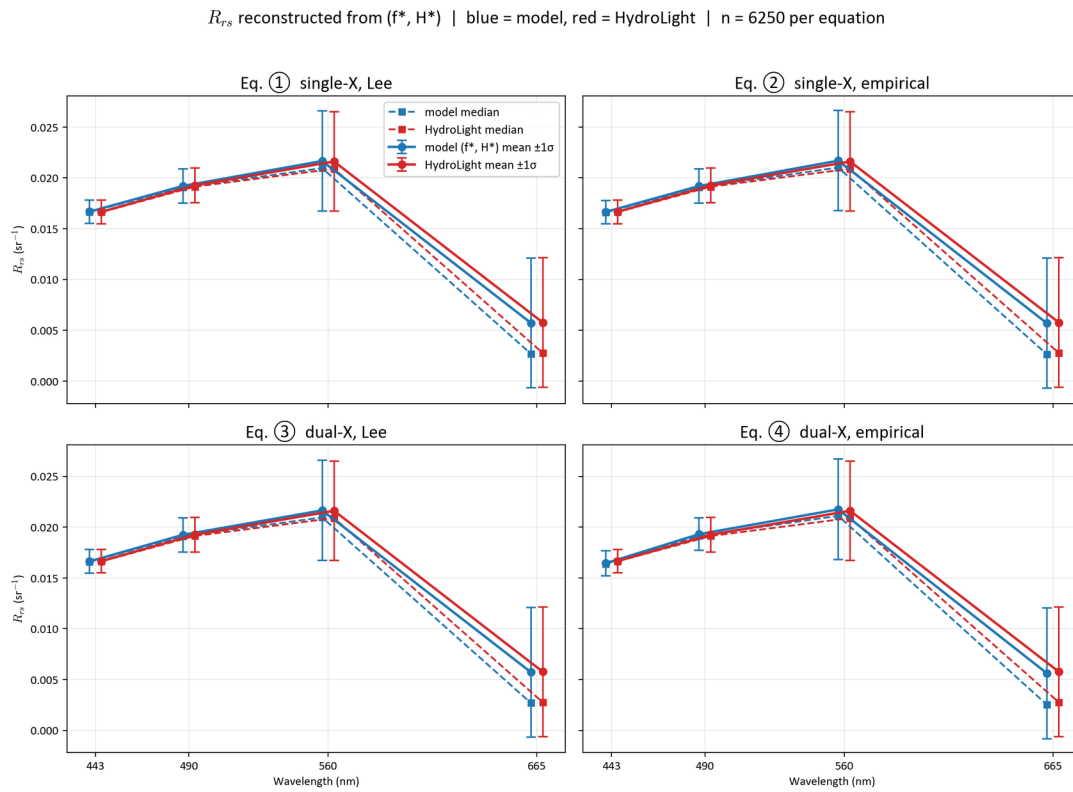


Figure 7. R_{rs} reconstructed from the retrieved mixing fractions and depth (blue) and HydroLight (red). Closer overlap means better reconstruction. $n = 6250$ per equation.

In Figure 7, ① and ③ almost overlap in blue and red. ② is also close. ④ has the best depth, but the spectrum is the most separated. This matches the R_{rs} RMSE in Table 6.

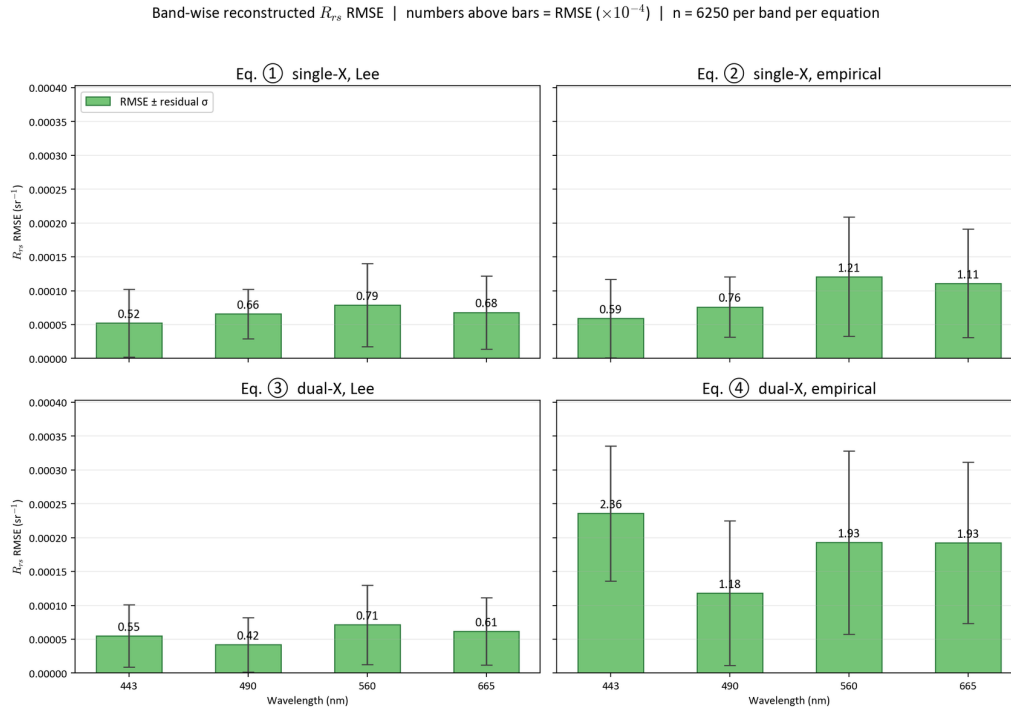


Figure 8. Band-wise reconstructed R_{rs} RMSE. Numbers above the bars are RMSE $\times 10^{-4}$. $n = 6250$ per band per equation.

Figure 8 is the same. The bars for ① and ③ are low, and ④ is high in all four bands. On a mixed bottom, depth and mixing fractions point to ② or ④; once reconstructed R_{rs} is included, ④ is weak.

Remaining limitation

Remaining limitation

The stopping rule has been decided. The tolerance number is kept at the SLSQP default 0.000001 (10^{-6}), and Cost is multiplied by 1 million (10^6) to match the scale of the numbers. The 1 million is the reciprocal of the square of the leftover R_{rs} error (about 0.001) when the true answer is inserted in Item 1. The tolerance was not increased.

The remaining limitation is the validation data. The mixed HydroLight set is a single grid of brown algae 80% + ooid sand 20%. On pure bottoms and other mixing fractions these 6,250-case numbers do not demonstrate the result.

Conclusion

Conclusion

The original method fixes the bottom and retrieves depth only with Cost (the sum of squared differences between observed R_{rs} and modelled R_{rs}). The three-band option was kept as a reference because, in Item 1, the relative error at 665 nm was large; it is not used in this Cost. The method used by the advisor selects the top K by the mean residual of representative Seeds and then jointly optimizes bottom mixing fractions and depth. This calculation keeps that procedure unchanged, and changes only Cost to the Item-4 sum of squares, and only the depth range to 0.001–6 m.

The validation data are a single HydroLight mixed-bottom grid of brown algae 80% + ooid sand 20%. The tolerance 0.000001 (10^{-6}) is the SLSQP default, so it was left as is. Cost is already below the tolerance, so depth and bottom are decided quickly from the starting values; the tolerance is left unchanged and the scale of the numbers is matched by $\text{Cost} \times 1 \text{ million}$ (10^6). The 1 million is the reciprocal of the square of the Item-1 forward R_{rs} error (about 0.001). The location of the minimum does not change. This is what brings the mixing fractions close to the truth. On a mixed bottom, depth is good for ② and ④, and reconstructed R_{rs} is better for ③ and ①. The stopping rule is $\text{Cost} \times 1 \text{ million} + \text{tolerance } 0.000001$. Other mixing fractions and pure bottoms are left to the next step.